TreemapMix: Dirichlet-Controlled Multi-Image Augmentation for Probability and Ordinal Supervision

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Abstract

Data augmentation improves visual recognition by exposing models to synthetic training examples that encourage generalization beyond the observed data. Mixed-sample methods extend this idea by combining multiple images. However, existing approaches either mix only pairs of images or rely on fixed small layouts, limiting control over source count, visible area proportions, and induced supervision. We propose TreemapMix, a distribution-controlled multi-image augmentation that samples source weights from a Dirichlet distribution and assigns them to area-proportional regions using a treemap partition. The resulting known region areas support two complementary objectives. TreemapMix with soft cross-entropy (SCE) treats areas as soft class-probability targets, while TreemapMix with a Plackett-Luce (PL) objective converts areas into ordinal supervision. On ImageNet-1K under a matched training protocol, TreemapMix-SCE achieves substantially lower calibration error than evaluated mixed-sample baselines, while TreemapMix-PL achieves the strongest Top-1 accuracy. Transfer experiments on object detection and instance segmentation suggest that TreemapMix-pretrained features remain useful beyond classification. These results show that controlled multi-image composition can expose the same region-area information as either probability-matching or rank-matching supervision. Code is available at github.com/Ejafa/treemapmix.

1 Introduction

Image classification is usually formulated as learning one target label from one image. Natural visual experience is more compositional: multiple objects often appear in the same field of view, and visual categories are often understood through comparisons among related sources of evidence. This suggests a complementary augmentation principle: construct training examples that expose several labeled sources jointly, while preserving explicit control over how much visual evidence each source contributes.

Mixed-sample data augmentation moves in this direction by composing examples and mixing their labels. Mixup [49] and CutMix [48] are widely used in modern image-classification recipes [26, 41, 42, 45], but they are primarily pairwise: Mixup globally interpolates two images, and CutMix replaces one region with a patch from another image. Multi-image methods such as RICAP [39] and Mosaic-style augmentation [3] show the value of composing several images, but typically use a fixed small number of sources and restricted layout families. Moreover, existing methods usually treat region area only as a soft-label weight, rather than as a controllable object that can shape both

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(a) None (b) Mixup (c) CutMix (d) RICAP (e) TreemapMix

Figure 1: Qualitative comparison of five augmentation strategies on matched source images using resize-and-center-crop preprocessing only. Mixup blends full images, CutMix pastes a random region, RICAP composes four tiles, and TreemapMix composes K rescaled source images within a randomized layout.

the visual composition and the supervision signal. Figure 1 shows a qualitative comparison of these augmentation patterns on matched source images.

We propose *TreemapMix*, a K-way image-mixing augmentation that composes multiple single-label images into a shared canvas while preserving the training sample count. TreemapMix samples region weights from a Dirichlet distribution and maps them to non-overlapping area-proportional regions using a squarified treemap layout [4]. The Dirichlet concentration parameter α controls the composition, ranging from dominant-region mixtures to more balanced multi-image mixtures. To prevent persistent label-to-area or label-to-rank bias, TreemapMix generates K cyclic views from each group of K source images, so that every source image is paired with every sampled region weight across the cyclic orbit. Area-preserving symmetries then increase spatial diversity without changing the target weights.

TreemapMix also exposes a supervision choice. Region area provides a natural but imperfect proxy for visual evidence: larger regions usually contain more pixels, more context, and more object parts, and are therefore often easier to recognize than smaller regions. This suggests an ordinal learning signal within a composed image, where labels assigned to larger regions should generally receive higher scores than labels assigned to smaller regions. However, area is not a calibrated measure of semantic evidence; a large region may contain mostly background, while a small region may contain a discriminative object part. We therefore study two complementary objectives: TreemapMix-SCE, which treats region areas as soft probability targets and optimizes soft cross-entropy, and TreemapMix-PL, which converts region areas into area-induced ordinal supervision by ranking the target classes and optimizing a Plackett-Luce listwise objective [31, 34, 46].

We evaluate TreemapMix on ImageNet-1K [10] under matched training recipes across multiple modern image-classification backbones. Compared with 13 baseline configurations, TreemapMix-PL achieves the strongest Top-1 accuracy across the evaluated ImageNet-1K backbones, while TreemapMix-SCE gives the lowest expected calibration error (ECE). The original NLL comparison also favors SCE, while Brier-score gains are smaller and mixed (Appendix F.1). Further diagnostics show that TreemapMix-PL improves alignment between logits and the area-induced ordering of composed regions, and COCO transfer experiments indicate that TreemapMix-pretrained representations remain useful for detection and instance segmentation [28].

Our contributions are:

i. We introduce TreemapMix, a Dirichlet-controlled K-way image-mixing augmentation that supports variable-source composition with explicit area-proportional treemap layouts while preserving the training sample count.

ii. We formulate two supervision objectives from the same composed image: area-proportional soft cross-entropy for probability matching and Plackett-Luce ranking for area-induced ordinal supervision.

iii. We analyze how the Dirichlet concentration parameter controls target variability, probability-matching supervision, and ranking emphasis.

iv. We provide a matched empirical evaluation on ImageNet-1K, including accuracy, calibration metrics, area-logit alignment analysis, and COCO detection and instance-segmentation transfer experiments.

2 Related Work

Standard augmentations preserve the original label, including region dropout methods such as Cutout and Random Erasing [11, 50] and policy-based methods such as AutoAugment, RandAugment, and AugMix [7, 8, 16]. Pairwise mixed-sample methods instead combine two examples and assign mixed targets: Mixup [49] interpolates images and labels, while CutMix [48] uses spatial replacement with area-based labels. Later methods refine pairwise composition or label estimation through resized patches, Fourier masks, grids, saliency, class activation maps, attention, or token-level mixing [1, 5, 14, 17, 23, 29, 35]. These methods improve label-preserving or two-source augmentation, whereas TreemapMix constructs multi-source examples with a controllable distribution over more than two visible labels.

Multi-image augmentation and label supervision: Multi-image augmentations compose more than two sources, including RICAP, which patches four crops with area-weighted labels [39], Mosaic augmentation in detection pipelines [3], Cut-thumbnail [47], StackMix [6], and DCutMix [19]. Other methods generalize mixing beyond spatial layouts, such as k-Mixup through optimal-transport interpolation [13] and MixMo through a multi-input multi-output architecture [36]. TreemapMix is closest to spatial multi-image methods such as RICAP and Mosaic, but samples a single coherent K-way area distribution from a Dirichlet distribution and realizes it as an area-proportional treemap layout; unlike k-Mixup and MixMo, it remains spatially explicit and uses a standard single-input model interface. Most spatial mixing methods treat visible area as a soft-label weight, although area may be an imperfect proxy for semantic evidence. Related work studies confidence and target behavior under mixup-style training through SmoothMixup [18], RegMixup [33], and RankMixup [32]. Plackett-Luce ranking has also been used for classification in Plackett–Luce Distillation (PLD) [2], but in a knowledge-distillation setting where a teacher model provides the class ranking. In contrast, TreemapMix induces the ranking directly from the mixed input: the Dirichlet-sampled region areas define the area-induced ordering over target classes. TreemapMix-SCE follows the area-weighted probability interpretation, while TreemapMix-PL converts the same areas into ordinal supervision with a Plackett-Luce ranking objective.

3 Method

TreemapMix is a K-way image-mixing augmentation that replaces each group of K source images with K cyclic mixed samples. Each mixed sample is supervised either by a soft probability-matching objective or by a ranking-based Plackett–Luce objective. We use integer $K \geq 1$ for the group size, C for the number of classes, and $\mathcal{G} = \{(x_k, y_k)\}_{k=1}^{K}$, where $y_k \in \{1, \ldots, C\}$, for a source group.

3.1 TreemapMix Augmentation

For each group, sample $w \sim \mathrm{Dirichlet}(\alpha, \ldots, \alpha)$, $\alpha > 0$, with $\sum_k w_k = 1$. These are slot weights, not fixed source-image weights. Smaller α favors dominant-region mixtures; larger α yields more balanced mixtures.

For the balanced-data protocol, TreemapMix first forms class-balanced rows from the training set and then splits each row into groups of size K. This encourages diverse multi-class mixtures and reduces the tendency of frequent classes to dominate mixed samples. When $K \leq C$, complete groups formed from a class-balanced row contain distinct labels by construction. ImageNet-LT instead uses natural-frequency grouping.

For stable layout construction and for ranking-based supervision, TreemapMix sorts the sampled weights in ascending order: $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_K, \lambda_i = w_{\tau(i)}$ (1). Here τ is a permutation of the source indices. The sorted weights define K layout slots. Slot i has area weight λ_i, with slot 1 the smallest-area slot and slot K the largest-area slot. Given the sorted weights, TreemapMix partitions the image canvas $\Omega = [0, W) \times [0, H)$ into half-open, non-overlapping rectangular boxes $\mathcal{B} = \{B_1, \ldots, B_K\}$ with $|B_i|/(HW) \approx \lambda_i$, such that $B_i \cap B_j = \emptyset$ for $i \neq j$ and $\bigcup_{i=1}^{K} B_i = \Omega$. We use a squarified treemap layout [4] to obtain area-proportional regions while favoring compact rectangles. This helps preserve useful semantic content inside each patch. The normalized treemap boxes are converted to integer pixel coordinates. If integer conversion yields zero-area boxes,

TreemapMix falls back to a stripe layout with approximate area matching; target weights remain unchanged.

Area-preserving spatial symmetries increase diversity without changing targets: the eight D_4 symmetries for square images, or identity, 180° rotation, and horizontal/vertical flips for rectangular images.

TreemapMix then generates K cyclic views from each group. Let slot i have weight λ_i, box B_i, and binary mask M_i. For cyclic view $s \in \{0, \dots, K-1\}$, define $\pi_s(i) = 1 + \big((i+s-1) \bmod K\big)$ (2). The s-th mixed image is $\tilde{x}^{(s)} = \sum_{i=1}^{K} M_i \odot \mathrm{Resize}(x_{\pi_s(i)}, B_i)$, with sparse label-weight target $\mathcal{T}^{(s)} = \{(y_{\pi_s(i)}, \lambda_i)\}_{i=1}^{K}$ (3). Here $\odot$ denotes elementwise multiplication, and Resize embeds the resized patch in the full canvas. Thus, in each cyclic view, the source image assigned to slot i receives the slot weight λ_i as its target weight.

All K cyclic views are retained for complete groups, avoiding enumeration of $K!$ assignments. Each source visits each slot once, so mean SCE target mass is m_c/K for source-label multiplicity m_c, uniform for distinct labels (Lemma 3). This prevents persistent source-to-slot bias, but does not remove prediction-dependent PL competition. Partial orbits lack the full balancing guarantee.

Appendix A gives pseudocode, remainder handling, geometry, and fallback details. Output count matches the constructed row stream. The sparse target supports soft probability supervision, or PL ranking for distinct-label groups.

3.2 Soft TreemapMix Cross-Entropy

Let $z = f_\theta(\tilde{x})$ be the model logits (class scores before softmax), with $p_c = \mathrm{softmax}(z)_c$. We write $\mathbf{1}[\cdot]$ for the indicator function. For a TreemapMix sample $\tilde{x}$ with sparse target $\mathcal{T} = \{(c_i, \eta_i)\}_{i=1}^{K}$, where $\eta_i \geq 0$ and $\sum_{i=1}^{K} \eta_i = 1$, and where c_i is the class assigned to slot i and η_i is its area weight, TreemapMix defines $q_c = \sum_{i=1}^{K} \eta_i \mathbf{1}[c_i = c], p_c = p_\theta(c \mid \tilde{x})$ (4). For cyclic view s, we have $c_i = y_{\pi_s(i)}$ and $\eta_i = \lambda_i$. If all source labels are distinct, then $q_{c_i} = \eta_i$. If a class appears multiple times in the group, its weights are accumulated. Thus q is a probability distribution over the C classes. We denote the cross-entropy objective of TreemapMix-SCE and the ranking objective of TreemapMix-PL by $\mathcal{L}_{\text{TM-CE}}$ and $\mathcal{L}_{\text{TM-PL}}$, respectively. We obtain the soft cross-entropy objective by minimizing the KL divergence [24] between q and p_θ. Since the entropy of q is independent of the model parameters, this is equivalent to minimizing

$$\mathcal{L}_{\text{TM-CE}} = -\sum_{c=1}^{C} q_c \log p_\theta(c \mid \tilde{x}) = -\sum_{i=1}^{K} \eta_i \log p_\theta(c_i \mid \tilde{x}). \tag{5}$$

For a mini-batch of size B, the average loss is $\mathcal{L}_{\text{TM-CE}} = -\frac{1}{B} \sum_{b=1}^{B} \sum_{i=1}^{K} \eta_{b,i} \log p_\theta(c_{b,i} \mid \tilde{x}_b)$ (6). Thus, TreemapMix-SCE treats the TreemapMix weights as area-proportional probability targets and encourages predicted probability mass to match the mixed-image area weights.

The per-sample logit gradient has the standard prediction-minus-target form: $\frac{\partial \mathcal{L}_{\text{TM-CE}}}{\partial z_c} = p_c - q_c$ (7). For the batch mean, the derivative with respect to each sample's logits additionally includes $1/B$. For complete distinct-label groups, the full cyclic orbit has exactly balanced mean targets, while a uniformly selected cyclic view retains Dirichlet variability. Appendix Lemma 1 derives its α-controlled variance under unfiltered sampling. Fixed rank positions instead follow Dirichlet order statistics. With logits fixed, the gradient $p_c - q_c$ has the same target-side variance.

3.3 TreemapMix Plackett–Luce Loss

As motivated in the introduction, area is an imperfect proxy for semantic evidence. The PL objective uses the same weights as an ordinal signal, encouraging larger-area target classes to rank above smaller-area classes rather than matching their probabilities exactly.

The PL objective assumes distinct target labels. Let $\mathcal{T} = \{(c_i, \eta_i)\}_{i=1}^{K}$ be a target with distinct c_i, and define a slot permutation σ with $r_\ell = c_{\sigma(\ell)}$ and class-associated weight $\rho_{r_\ell} = \eta_{\sigma(\ell)}$. Order these weights descending: $\rho_{r_1} \geq \rho_{r_2} \geq \cdots \geq \rho_{r_K}, r_1 \succ r_2 \succ \cdots \succ r_K$ (8). Let $\mathcal{U} = \{1, \dots, C\} \setminus \{r_1, \dots, r_K\}$ be the set of classes absent from the TreemapMix target, and let $p_{\mathcal{U}} = 1 - \sum_{j=1}^{K} p_{r_j}$

be their total predicted probability mass. The ranking likelihood selects classes sequentially from largest to smallest target weight, normalizing each choice over the remaining classes. Following the Plackett–Luce model [31, 34] and listwise maximum-likelihood (ListMLE) objectives [46], the descending TreemapMix-PL loss is

$$\mathcal{L}_{\text{TM-PL}} = -\sum_{\ell=1}^{K} \rho_{r_\ell} \log \frac{p_\theta(r_\ell \mid \tilde{x})}{p_{\mathcal{U}} + \sum_{j=\ell}^{K} p_\theta(r_j \mid \tilde{x})} \tag{9}$$

Similar to position-aware listwise objectives such as Position-Aware ListMLE [25] and PLD [2], TreemapMix-PL weights each ranking term by its target importance; this weighted surrogate is not itself a normalized ranking likelihood. TreemapMix-SCE and TreemapMix-PL therefore use the same TreemapMix target in complementary ways: TreemapMix-SCE seeks $p_\theta(c \mid \tilde{x}) \approx q_c$, whereas TreemapMix-PL seeks $r_1 \succ r_2 \succ \cdots \succ r_K$.

For implementation, we use the ascending order. Let $a_i = r_{K+1-i}$, $\gamma_i = \rho_{a_i}$, and $\gamma_1 \leq \gamma_2 \leq \cdots \leq \gamma_K$. Thus, a_1 is the weakest ranked target class and a_K is the strongest. Define $D_i = p_{\mathcal{U}} + \sum_{j=1}^{i} p_{a_j} = 1 - \sum_{j=i+1}^{K} p_{a_j}$ (10). Then the loss becomes $\mathcal{L}_{\text{TM-PL}} = \sum_{i=1}^{K} \gamma_i \left[\log D_i - \log p_{a_i}\right]$ (11). If $T_i = \{a_{i+1}, \ldots, a_K\}$ (12) denotes the set of classes stronger than a_i, then the logit gradient is

$$\frac{\partial \mathcal{L}_{\text{TM-PL}}}{\partial z_c} = \sum_{i=1}^{K} \gamma_i \left[\frac{p_c\big(1 - \mathbf{1}[c \in T_i]\big)}{D_i} - \mathbf{1}[c = a_i] \right] \tag{13}$$

At the logit level, this gradient suppresses unranked classes, promotes the strongest ranked class, and regulates intermediate ranked classes through competition with stronger classes. The full likelihood derivation, ascending reindexing, and gradient proof are given in Appendix C.

For distinct target labels, the ascending PL weights satisfy $\gamma_i = \lambda_i$. The sorted Dirichlet weights determine the importance assigned to each rank position. Appendix Lemma 2 in Appendix C characterizes how α controls this ranking emphasis through the order statistics of the sampled Dirichlet weights. This analysis holds logits and rank assignments fixed and assumes unfiltered Dirichlet sampling; filtering changes the order-statistic moments. As α increases, the sorted weights concentrate near $1/K$, so in the distinct-label case the loss approaches $1/K$ times an unweighted top-K ListMLE-style objective. Equal-weight ties retain source-slot order.

For completeness, we also evaluate a convex combination of TreemapMix-SCE and TreemapMix-PL. Let $\beta \in [0, 1]$ denote the interpolation coefficient, where $\beta = 1$ recovers TreemapMix-SCE and $\beta = 0$ recovers TreemapMix-PL. We define $\mathcal{L}_{\text{TM-Mix}} = \beta \mathcal{L}_{\text{TM-CE}} + (1 - \beta)\mathcal{L}_{\text{TM-PL}}$ (14). Using the ascending PL notation from Section 3.3, this can be written as $\mathcal{L}_{\text{TM-Mix}} = \sum_{i=1}^{K} \gamma_i \left[(1 - \beta) \log D_i - \log p_{a_i}\right]$ (15). This objective is not intended as a separate contribution; we include it to test whether combining probability matching and rank matching improves the accuracy–calibration trade-off.

3.4 Dirichlet-Controlled Supervision

TreemapMix supports a fixed Dirichlet concentration or a schedule over training progress. We vary α between endpoints $\alpha_{\min}$ and $\alpha_{\max}$ using a linear or cosine schedule. A forward schedule increases α, shifting from dominant-region mixtures toward more balanced mixtures; a backward schedule decreases it. Cyclic permutation prevents a fixed source label from remaining tied to one slot as the schedule changes.

4 Empirical Evaluation

Protocol. Our main ImageNet-1K study uses four compact ViT/EVA02-family backbones in `timm` [12], two NVIDIA H20 GPUs per run, global batch size 1024, and a matched 110-epoch recipe. Unless specified otherwise, results are mean $\pm$ one sample standard deviation over seeds 42, 43, and 44. Accuracy is Top-1 percent; ECE is the sample-weighted absolute accuracy–confidence gap across 15 equal-width bins, in percentage points. TreemapMix uses $K = 4$ for SCE/mixed, $K = 6$ for PL, and cosine $\alpha : 0.1 \to 0.5$; mixed uses $\beta = 0.5$. Settings selected on ViT-Wee are fixed across backbones. Appendix E.1 gives compute, recipes, and hyperparameters; Appendix E.4 gives mathematical loss references.

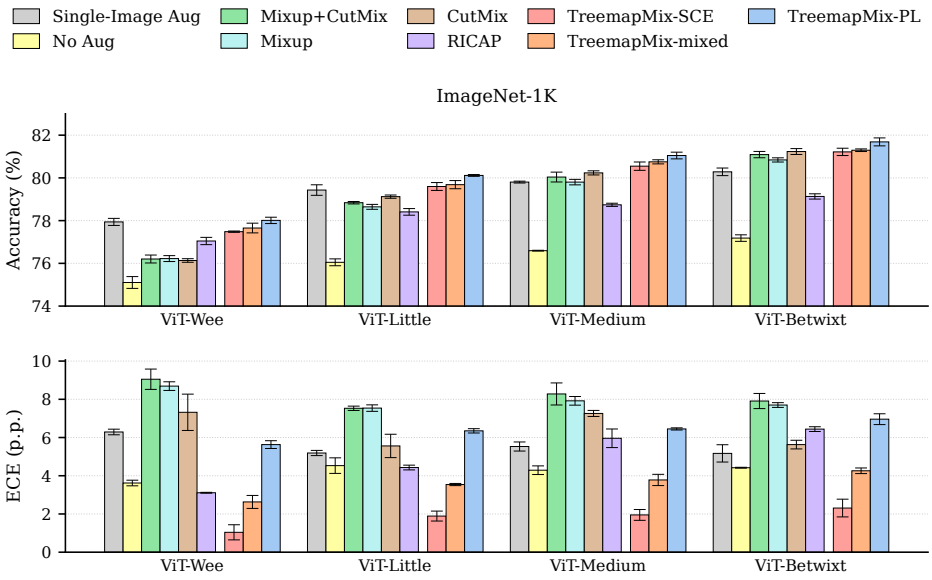


Figure 2: ImageNet-1K main comparison under the matched 110-epoch protocol. The original nine configurations are shown. Bars show means over three seeds and error bars denote ± 1 sample standard deviation. ECE uses 15 bins and is reported in percentage points.

ImageNet-1K. Tables 1 and 2 report all 16 configurations, including the additional review-time baselines. FMix, GridMix, ResizeMix, SaliencyMix, SmoothMix, TokenMix, and TLA are OpenMixup-based reimplementations [26], each selected from eight configurations on ViT-Wee. PL has the highest mean accuracy on every backbone, whereas SCE has the lowest ECE. The strongest baseline varies: single-image augmentation on Wee, SaliencyMix on Little, and ResizeMix on Medium/Betwixt. The mixed loss does not consistently improve on both objectives. Figure 2 shows the original nine configurations; NLL/Brier results appear in Appendix F.1.

Table 1: ImageNet-1K Top-1 accuracy (%) under the matched 110-epoch protocol. Mean $\pm$ sample standard deviation over three seeds; bold marks the best mean per backbone.

Method	ViT-Wee ↑	ViT-Little ↑	ViT-Medium ↑	ViT-Betwixt ↑
No Aug	75.10 ± 0.24	76.05 ± 0.14	76.59 ± 0.02	77.18 ± 0.13
Single-Image Aug	77.94 ± 0.15	79.43 ± 0.22	79.80 ± 0.04	80.28 ± 0.16
Mixup+CutMix	76.20 ± 0.16	78.84 ± 0.05	80.04 ± 0.20	81.09 ± 0.13
Mixup	76.22 ± 0.12	78.64 ± 0.10	79.80 ± 0.11	80.84 ± 0.09
CutMix	76.13 ± 0.07	79.12 ± 0.07	80.23 ± 0.09	81.23 ± 0.12
RICAP	77.05 ± 0.15	78.41 ± 0.14	78.74 ± 0.07	79.13 ± 0.11
FMix	76.73 ± 0.10	79.40 ± 0.06	80.30 ± 0.07	81.12 ± 0.10
GridMix	76.39 ± 0.17	78.72 ± 0.05	79.50 ± 0.09	80.33 ± 0.18
ResizeMix	76.96 ± 0.17	79.51 ± 0.15	80.53 ± 0.09	81.44 ± 0.13
SaliencyMix	76.93 ± 0.11	79.53 ± 0.07	80.35 ± 0.12	81.22 ± 0.06
SmoothMix	76.22 ± 0.22	79.27 ± 0.09	80.26 ± 0.08	81.18 ± 0.05
TokenMix	76.43 ± 0.10	79.19 ± 0.13	80.27 ± 0.13	81.18 ± 0.10
TLA	76.81 ± 0.19	79.47 ± 0.16	80.46 ± 0.03	81.32 ± 0.13
TreemapMix-SCE	77.48 ± 0.03	79.60 ± 0.16	80.55 ± 0.17	81.22 ± 0.15
TreemapMix-mixed	77.65 ± 0.20	79.68 ± 0.17	80.75 ± 0.09	81.30 ± 0.05
TreemapMix-PL	$\mathbf{78.01 \pm 0.13}$	$\mathbf{80.12 \pm 0.04}$	$\mathbf{81.05 \pm 0.14}$	$\mathbf{81.69 \pm 0.16}$

Other datasets and architectures. Tables 3 and 4 contain the complete CIFAR-100 comparison at 256-pixel resolution under 110 epochs including 10 warmup epochs. SCE has the lowest ECE on all backbones; PL has the highest accuracy on three, while ResizeMix leads on Little. Table 5 retains the accuracy/calibration separation on ConvNeXtV2-Nano, DeiT-Tiny, and Swin-Tiny. Small PL–SCE accuracy gaps on DeiT/Swin are not claimed statistically significant.

On ImageNet-LT [40], the natural-mixed scheduler preserves every original sample and the long-tailed class frequencies (Appendix F.2). Under the matched protocol, PL improves accuracy over

Table 2: ImageNet-1K ECE (15 bins, percentage points). Same runs as Table 1; bold marks the lowest mean per backbone.

Method	ViT-Wee ↓	ViT-Little ↓	ViT-Medium ↓	ViT-Betwixt ↓
No Aug	3.62 ± 0.13	4.53 ± 0.36	4.29 ± 0.20	4.42 ± 0.02
Single-Image Aug	6.29 ± 0.13	5.19 ± 0.12	5.53 ± 0.21	5.17 ± 0.40
Mixup+CutMix	9.05 ± 0.47	7.53 ± 0.10	8.28 ± 0.51	7.91 ± 0.35
Mixup	8.69 ± 0.20	7.54 ± 0.15	7.92 ± 0.20	7.70 ± 0.11
CutMix	7.32 ± 0.84	5.56 ± 0.54	7.26 ± 0.14	5.63 ± 0.20
RICAP	3.11 ± 0.02	4.43 ± 0.11	5.96 ± 0.43	6.44 ± 0.11
FMix	6.40 ± 0.19	5.01 ± 0.10	5.44 ± 0.26	5.12 ± 0.26
GridMix	7.04 ± 0.15	5.72 ± 0.17	5.66 ± 0.54	5.48 ± 0.32
ResizeMix	7.18 ± 0.28	5.40 ± 0.74	6.06 ± 0.17	5.68 ± 0.55
SaliencyMix	6.41 ± 0.43	5.09 ± 0.09	5.39 ± 0.16	4.57 ± 0.29
SmoothMix	8.52 ± 0.06	6.93 ± 0.18	7.54 ± 0.06	6.88 ± 0.21
TokenMix	7.60 ± 0.65	5.86 ± 0.65	5.89 ± 0.75	5.53 ± 0.59
TLA	5.74 ± 0.36	4.68 ± 0.03	4.85 ± 0.17	4.53 ± 0.36
TreemapMix-SCE	$\mathbf{1.04 \pm 0.35}$	$\mathbf{1.89 \pm 0.23}$	$\mathbf{1.95 \pm 0.25}$	$\mathbf{2.31 \pm 0.41}$
TreemapMix-mixed	2.63 ± 0.30	3.54 ± 0.05	3.78 ± 0.26	4.26 ± 0.13
TreemapMix-PL	5.63 ± 0.18	6.35 ± 0.10	6.45 ± 0.05	6.96 ± 0.25

Table 3: CIFAR-100 Top-1 accuracy (%) at 256 pixels, 110 epochs including 10 warmup epochs. Mean ± sample standard deviation over three seeds; bold marks the best mean per backbone.

Method	ViT-Wee ↑	ViT-Little ↑	ViT-Medium ↑	ViT-Betwixt ↑
No Aug	65.05 ± 0.44	63.69 ± 0.81	60.72 ± 0.54	60.70 ± 0.35
Single-Image Aug	71.50 ± 0.09	71.97 ± 0.45	72.56 ± 0.35	72.16 ± 0.06
Mixup+CutMix	74.15 ± 0.47	74.96 ± 0.64	75.81 ± 0.24	75.83 ± 0.16
Mixup	73.13 ± 0.10	73.46 ± 0.34	74.18 ± 0.18	74.02 ± 0.18
CutMix	73.39 ± 0.51	74.05 ± 0.16	75.21 ± 0.22	74.92 ± 0.83
FMix	74.58 ± 0.37	74.84 ± 0.22	75.52 ± 0.22	75.40 ± 0.07
GridMix	70.58 ± 0.16	70.34 ± 0.27	71.34 ± 0.84	71.48 ± 0.22
ResizeMix	74.99 ± 0.22	$\mathbf{75.54 \pm 0.34}$	75.89 ± 0.05	76.18 ± 0.12
SaliencyMix	74.60 ± 0.14	74.64 ± 0.29	75.43 ± 0.40	75.60 ± 0.28
SmoothMix	74.42 ± 0.11	75.00 ± 0.36	75.38 ± 0.23	75.74 ± 0.28
TLA	75.42 ± 0.13	75.27 ± 0.13	75.93 ± 0.02	75.96 ± 0.18
TokenMix	74.75 ± 0.28	75.18 ± 0.24	75.36 ± 0.25	75.82 ± 0.44
RICAP	71.71 ± 0.35	72.07 ± 0.38	72.90 ± 0.27	72.26 ± 0.26
TreemapMix-SCE	75.29 ± 0.11	75.20 ± 0.36	75.72 ± 0.11	75.73 ± 0.09
TreemapMix-mixed	75.10 ± 0.37	75.38 ± 0.24	75.89 ± 0.15	75.70 ± 0.39
TreemapMix-PL	$\mathbf{75.72 \pm 0.29}$	75.42 ± 0.45	$\mathbf{75.97 \pm 0.29}$	$\mathbf{76.39 \pm 0.35}$

Table 4: CIFAR-100 ECE (15 bins, percentage points). Same runs as Table 3; bold marks the lowest mean per backbone.

Method	ViT-Wee ↓	ViT-Little ↓	ViT-Medium ↓	ViT-Betwixt ↓
No Aug	7.64 ± 0.20	6.19 ± 0.39	4.61 ± 0.32	5.01 ± 0.08
Single-Image Aug	3.16 ± 0.17	3.75 ± 0.24	3.44 ± 0.22	3.90 ± 0.30
Mixup+CutMix	14.43 ± 0.28	14.16 ± 0.50	14.07 ± 0.19	13.93 ± 0.22
Mixup	12.67 ± 0.18	12.24 ± 0.24	12.23 ± 0.05	12.14 ± 0.24
CutMix	11.74 ± 0.02	10.85 ± 0.37	10.35 ± 0.09	10.49 ± 0.28
FMix	5.93 ± 0.14	5.57 ± 0.18	5.19 ± 0.14	4.86 ± 0.22
GridMix	3.67 ± 0.21	3.30 ± 0.26	3.68 ± 0.33	3.69 ± 0.22
ResizeMix	7.87 ± 0.66	7.41 ± 0.76	7.50 ± 0.34	7.52 ± 0.26
SaliencyMix	6.01 ± 0.49	5.35 ± 0.52	5.64 ± 0.63	5.10 ± 0.16
SmoothMix	11.58 ± 0.10	11.50 ± 0.25	11.11 ± 0.12	11.29 ± 0.33
TLA	7.14 ± 0.10	6.93 ± 0.12	6.54 ± 0.29	6.44 ± 0.31
TokenMix	10.47 ± 0.47	10.35 ± 0.37	9.89 ± 0.17	9.64 ± 0.45
RICAP	10.17 ± 0.11	10.81 ± 0.44	9.87 ± 0.14	9.99 ± 0.42
TreemapMix-SCE	$\mathbf{1.62 \pm 0.20}$	$\mathbf{1.62 \pm 0.20}$	$\mathbf{1.87 \pm 0.17}$	$\mathbf{1.79 \pm 0.08}$
TreemapMix-mixed	3.81 ± 0.20	4.60 ± 0.35	4.55 ± 0.18	4.85 ± 0.27
TreemapMix-PL	9.70 ± 0.32	10.83 ± 0.14	10.55 ± 0.11	10.41 ± 0.40

Table 5: Cross-architecture ImageNet-1K results under 110 epochs. Mean ± sample standard deviation over three seeds; accuracy in percent and ECE in percentage points (15 bins). Bold marks the best mean per metric.

	ConvNeXtV2-Nano		DeiT-Tiny		Swin-Tiny	
Method	Acc. ↑	ECE ↓	Acc. ↑	ECE ↓	Acc. ↑	ECE ↓
Mixup+CutMix	79.79 ± 0.11	9.48 ± 0.13	68.86 ± 0.28	12.34 ± 0.27	79.62 ± 0.06	7.50 ± 0.13
RICAP	79.08 ± 0.10	5.95 ± 0.17	71.71 ± 0.04	2.04 ± 0.28	78.68 ± 0.07	7.78 ± 0.05
TreemapMix-SCE	80.01 ± 0.09	$\mathbf{2.74 \pm 0.27}$	71.78 ± 0.26	$\mathbf{1.32 \pm 0.26}$	80.55 ± 0.09	$\mathbf{2.52 \pm 0.11}$
TreemapMix-PL	$\mathbf{80.45 \pm 0.10}$	6.82 ± 0.07	$\mathbf{71.81 \pm 0.07}$	2.71 ± 0.15	$\mathbf{80.57 \pm 0.12}$	6.58 ± 0.06

Mixup+CutMix by 1.40, 1.02, 1.31, and 1.82 points; SCE's calibration advantage does not persist (Table 6).

Table 6: ImageNet-LT accuracy and ECE under the matched 110-epoch protocol. Mean ± sample standard deviation over three seeds; ECE uses 15 bins. Bold marks the best mean per metric.

Method	ViT-Wee	ViT-Little	ViT-Medium	ViT-Betwixt
	Top-1 accuracy (%) ↑			
Mixup+CutMix	41.72 ± 0.08	41.67 ± 0.43	40.95 ± 0.07	39.78 ± 0.21
RICAP	35.70 ± 0.15	35.39 ± 0.00	34.35 ± 0.02	31.44 ± 3.37
TreemapMix-SCE	41.44 ± 0.40	41.09 ± 0.40	39.48 ± 0.13	38.67 ± 0.27
TreemapMix-PL	$\mathbf{43.12 \pm 0.06}$	$\mathbf{42.69 \pm 0.07}$	$\mathbf{42.26 \pm 0.07}$	$\mathbf{41.60 \pm 0.17}$
	ECE (percentage points) ↓			
Mixup+CutMix	$\mathbf{0.58 \pm 0.14}$	$\mathbf{0.83 \pm 0.09}$	$\mathbf{2.14 \pm 0.61}$	$\mathbf{3.31 \pm 0.96}$
RICAP	28.50 ± 0.26	31.86 ± 0.21	33.12 ± 0.32	30.89 ± 2.81
TreemapMix-SCE	10.65 ± 0.04	13.00 ± 0.56	11.85 ± 0.33	11.35 ± 0.62
TreemapMix-PL	17.90 ± 0.63	25.63 ± 0.38	23.62 ± 0.74	25.45 ± 0.79

Computational performance. Table 7 reports every field of the review-time loader benchmark. Relative to single-image augmentation, SCE ($K = 4$) reduces throughput by 19.0% and PL ($K = 6$) by 25.6%, with increased host memory. These CPU-side measurements exclude model computation and do not establish end-to-end training overhead; prefetching can overlap preparation with GPU work (Appendix E.5).

Table 7: Complete loader benchmark: five runs (seeds 42–46), batch size 128, eight workers, first batch, 20 warmup and 100 measured batches. Reduction is $100(1 - T/T_{\text{single}})$; ΔRSS uses unrounded means. Memory is GiB (2^{30} bytes). Total time covers all consumed batches; throughput uses measured batches only.

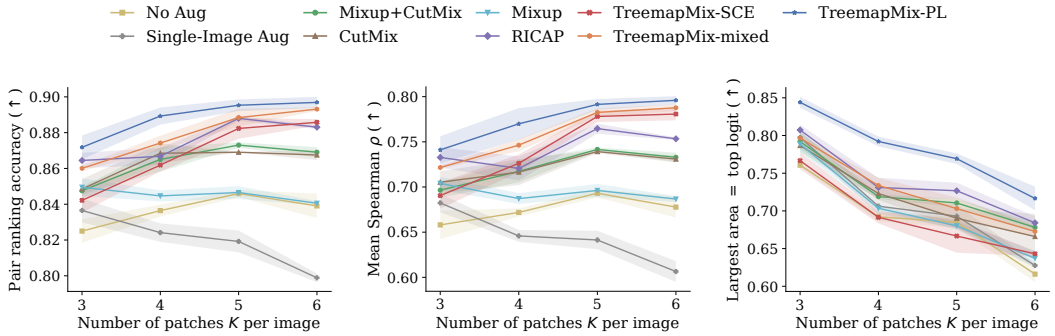
Method	K	Throughput (images/s)	Throughput reduction	Total loader time (s)
No Aug	1	1670.4 ± 30.2	-30.2%	11.43 ± 0.13
Single-Image Aug	1	1282.8 ± 35.1	0.0%	14.35 ± 0.27
Mixup+CutMix	2	1223.4 ± 29.7	4.6%	15.13 ± 0.36
TreemapMix-SCE	4	1039.4 ± 37.0	19.0%	19.83 ± 0.46
TreemapMix-PL	6	954.3 ± 33.0	25.6%	21.30 ± 0.36
RICAP	4	364.8 ± 5.9	71.6%	44.00 ± 0.47

Method	Peak RSS (GiB)	ΔRSS (GiB)	Peak USS (GiB)
No Aug	28.49 ± 0.07	$+0.11$	17.82 ± 0.06
Single-Image Aug	28.38 ± 0.16	$+0.00$	17.76 ± 0.17
Mixup+CutMix	28.46 ± 0.12	$+0.08$	17.82 ± 0.15
TreemapMix-SCE	30.85 ± 0.34	$+2.47$	20.16 ± 0.28
TreemapMix-PL	32.41 ± 0.52	$+4.04$	21.74 ± 0.49
RICAP	36.82 ± 0.14	$+8.44$	25.18 ± 0.13

Robustness and longer training. The 320-epoch Wee/Little comparison (20 warmup epochs followed by 300 regular epochs) retains PL's highest mean accuracy and SCE's lowest ECE, with smaller PL gains over Mixup+CutMix of 0.91/0.21 points (Table 8). It reuses short-schedule settings.

Table 8: ImageNet-1K long-horizon results for ViT-Wee and ViT-Little under a 320-epoch schedule. Accuracy is reported in percent, and ECE in percentage points using 15 bins. Values are mean ± one sample standard deviation over seeds 42, 43, and 44. Bold marks the best mean for each model and metric.

	ViT-Wee		ViT-Little	
Method	Acc. ↑	ECE ↓	Acc. ↑	ECE ↓
Single-Image Aug	80.37 ± 0.08	3.89 ± 0.16	81.12 ± 0.04	3.39 ± 0.06
Mixup+CutMix	79.73 ± 0.15	6.87 ± 0.64	81.74 ± 0.09	5.47 ± 0.52
TreemapMix-SCE	80.41 ± 0.04	$\mathbf{3.23 \pm 0.49}$	81.58 ± 0.33	$\mathbf{2.77 \pm 1.42}$
TreemapMix-PL	$\mathbf{80.64 \pm 0.16}$	6.94 ± 0.14	$\mathbf{81.95 \pm 0.34}$	7.44 ± 0.54



(a) Pairwise rank accuracy (b) Area-logit Spearman correlation (c) Largest-area top-class accuracy

Figure 3: Area-logit alignment diagnostics on the held-out composed diagnostic set for ViT-Betwixt checkpoints from the main ImageNet-1K study. Lines show means over three seeds and shaded bands denote ± 1 sample standard deviation: pairwise rank accuracy (left), area-logit Spearman correlation (center), and largest-area top-class accuracy (right).

Single-seed optimizer and (K, α) studies appear in Appendices F.5 and F.6. The latter supports small-to-medium K and small α in the recommended regimes, without isolating filtering or scheduling effects.

5 Diagnostic and Transfer Analysis

We examine whether logits reflect the known region-area ordering and whether pretrained features transfer to COCO detection and instance segmentation. Hyperparameter sweeps and qualitative attributions appear in Appendices F.6 and F.7.

Compositional Logit Alignment: We evaluate the three-seed ViT-Betwixt checkpoints on 500 held-out mixtures for each $K \in \{3, 4, 5, 6\}$. Figure 3 measures pairwise rank accuracy, area–logit Spearman correlation, and largest-area top-class accuracy among present classes. Appendix E.2 specifies the pair threshold and tie handling.

TreemapMix-PL gives the highest pairwise rank accuracy, area-logit Spearman correlation, and largest-area top-class accuracy in Figure 3. RICAP is competitive across all three metrics and outperforms TreemapMix-SCE on Spearman correlation and largest-area top-class accuracy, despite also using area-weighted soft targets. These results suggest that the ranking objective contributes to area-logit alignment beyond the effect of compositional augmentation alone.

Feature Transfer to Instance Segmentation: We fine-tune ViT-Wee and ViT-Betwixt ImageNet-1K checkpoints with a ViTDet-style adapter and Mask R-CNN [15, 27] on COCO for 30 epochs at 256×256 resolution. Mixup+CutMix and RICAP provide the comparison baselines. Table 9 reports box/mask AP averaged over three transferred seeds; Appendix E.3 gives the full setup and AP50 results (Table 14).

Both TreemapMix variants improve over Mixup+CutMix and RICAP on box AP and mask AP for both backbones, with TreemapMix-PL achieving the highest scores. These results show that

Table 9: COCO transfer results after ViTDet-style Mask R-CNN fine-tuning from the main ImageNet-1K experiment checkpoints with three seeds per method. Values are mean ± standard deviation over these three transferred seeds; higher is better. Box AP evaluates bounding boxes and mask AP evaluates instance masks. AP is averaged over IoU thresholds from 0.50 to 0.95 in increments of 0.05.

	ViT-Wee		ViT-Betwixt	
Method	Box AP ↑	Mask AP ↑	Box AP ↑	Mask AP ↑
Mixup+CutMix	23.66 ± 0.01	20.34 ± 0.10	29.04 ± 0.07	25.25 ± 0.10
RICAP	23.08 ± 0.17	19.82 ± 0.07	27.86 ± 0.08	24.21 ± 0.06
TreemapMix-SCE	24.00 ± 0.17	20.55 ± 0.20	29.44 ± 0.16	25.56 ± 0.14
TreemapMix-PL	**24.32 ± 0.12**	**20.71 ± 0.04**	**29.71 ± 0.09**	**25.75 ± 0.10**

TreemapMix-pretrained checkpoints transfer favorably to COCO detection and instance segmentation under this fine-tuning protocol. However, they should not be interpreted as proving that the area-logit alignment mechanism causes the downstream gains, since the transfer experiment compares pretrained checkpoints rather than isolating individual representation properties.

6 Conclusion and Limitations

We introduced TreemapMix, a Dirichlet-controlled K-way augmentation with area-proportional treemap layouts. Cyclic views preserve sample count and, for complete groups, balance each source's exposure to region sizes and ranks. One composition supports area-weighted soft cross-entropy for probability matching and Plackett–Luce ranking for ordinal supervision.

Under matched ImageNet-1K recipes, SCE gives the strongest calibration, while PL gives the strongest Top-1 accuracy and area-logit alignment. The pattern extends to additional architectures and largely to CIFAR-100. On ImageNet-LT, PL improves accuracy but SCE's calibration advantage does not persist. COCO transfer supports useful detection and instance-segmentation representations.

Limitations. Our comparisons use matched training recipes and do not establish the best performance attainable with a separately optimized recipe for every method. The long-horizon study covers two backbones and reuses TreemapMix settings selected for the shorter schedule; its smaller margins limit claims about benefits at convergence. Region area is a weak cue for semantic evidence: a large region may contain mostly background. The area–logit diagnostics do not measure classification degradation on a controlled subset of such failure cases. We also do not isolate the empirical contributions of cyclic permutation and class-balanced grouping with dedicated ablations. The loader benchmark measures CPU-side throughput and host memory, rather than end-to-end training time or GPU memory. Finally, SCE's calibration advantage does not persist on ImageNet-LT, so the balanced-data findings should not be assumed to hold under a strongly skewed class prior. Our PL likelihood proofs assume distinct classes, and exact Dirichlet moments describe the unfiltered sampler; Appendix A distinguishes these assumptions from implementation behavior.

These results motivate adaptive grouping, automatic schedules for K and α, and extensions to contrastive or self-supervised learning.

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A TreemapMix Augmentation Details

This appendix provides implementation details omitted from the main Method section. We use the notation introduced in Section 3.

Algorithm 1 TreemapMix augmentation

Require: dataset $\mathcal{D}$, group size K, concentration $\alpha > 0$, image size $H \times W$, with $1 \le K \le \min(C, W)$ for this balanced-row protocol

build class-balanced rows and split them into groups of size K
for each group $\mathcal{G} = \{(x_k, y_k)\}_{k=1}^{K}$ **do**
  sample $w \sim \mathrm{Dirichlet}(\alpha, \dots, \alpha)$
  sort weights into $\lambda_1 \le \cdots \le \lambda_K$
  construct boxes $B_1, \dots, B_K$ with $|B_i|/(HW) \approx \lambda_i$
  apply a random valid symmetry transform to the boxes
  for $s = 0, \dots, K-1$ **do**
    assign each source image $x_{\pi_s(i)}$ to slot i, for $i = 1, \dots, K$
    render $\tilde{x}^{(s)} = \sum_i M_i \odot \mathrm{Resize}(x_{\pi_s(i)}, B_i)$
    output $\big(\tilde{x}^{(s)}, \{(y_{\pi_s(i)}, \lambda_i)\}_{i=1}^{K}\big)$
  end for
end for

A.1 Class-balanced row construction

Let the training dataset be $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^{N_{\mathrm{data}}}$, where $y_n \in \{1, \dots, C\}$, and let $\mathcal{I}_c = \{n : y_n = c\}$ be the set of dataset indices belonging to class c. A class-balanced row is constructed by sampling one example from each class. For imbalanced datasets, minority classes may be sampled with replacement, while majority classes may be subsampled across rows, so that every class contributes equally to each row.

Each row is shuffled and split into groups of size K. When $K \le C$, complete groups formed from a class-balanced row contain distinct labels by construction, because each class appears at most once in the row. This encourages diverse multi-class mixtures and avoids repeated target classes in most TreemapMix groups.

A.2 Incomplete groups and row-size preservation

Suppose a row contains N_{row} samples. Write $N_{\mathrm{row}} = K \left\lfloor \frac{N_{\mathrm{row}}}{K} \right\rfloor + r$, where $0 \le r < K$. TreemapMix processes the first $K \left\lfloor \frac{N_{\mathrm{row}}}{K} \right\rfloor$ samples as complete groups of size K. If $r > 0$, the remaining r samples are padded with $K - r$ samples drawn without replacement from earlier entries in the same row, and only the first r cyclic outputs are retained. Padding from disjoint earlier entries in a one-example-per-class row preserves distinct labels. Natural-frequency rows can already contain repeated classes; rows shorter than K are passed through unmixed. SCE accumulates repeated-class weights. The historical LabelMix PL implementation instead processes repeated entries separately, with a clamped residual mass. This is an implementation-level surrogate outside the distinct-class PL likelihood and gradient proofs below; those results do not justify repeated-entry ranking.

TreemapMix generates K cyclic views from the temporary group but retains only the first r outputs. Therefore, the total number of final TreemapMix samples contributed by the row is $K \left\lfloor \frac{N_{\mathrm{row}}}{K} \right\rfloor + r = N_{\mathrm{row}}$. Thus, output count matches the constructed row stream. Retaining only $r < K$ views does not provide the full-orbit balancing guarantee.

A.3 Integer box conversion and layout fallback

The squarified treemap is first computed in normalized coordinates. Let $\bar{B}_i = (\bar{x}_i, \bar{y}_i, \bar{w}_i, \bar{h}_i)$ denote the normalized rectangle for slot i. We convert it to an integer pixel box $B_i = (x_i^{(0)}, y_i^{(0)}, x_i^{(1)}, y_i^{(1)})$. Because independent rounding can introduce gaps or overlaps, shared boundaries are snapped consistently. Using half-open pixel boxes, a validated conversion satisfies $B_i \cap B_j = \emptyset$ for $i \ne j$ and $\bigcup_{i=1}^{K} B_i = [0, W) \times [0, H)$.

A converted layout is considered valid if every box has positive area. Equivalently, for integer boxes, every box must have width and height at least one pixel. If the treemap layout fails this validity check, TreemapMix falls back to a stripe layout. The stripe layout partitions the canvas along one axis according to the sorted weights $\lambda_1, \ldots, \lambda_K$, then applies the same integer-boundary snapping procedure. For vertical stripes, $K \leq W$ and $H \geq 1$ permit a positive-area full-canvas tiling. Boundaries are adjusted to give each stripe at least one pixel. Consequently, actual area fractions only approximate the sampled weights; minimum-width adjustments can exceed nearest-integer rounding error. Target weights remain the sampled weights.

A.4 Symmetry transformations

For square images, TreemapMix samples from the dihedral group

$$D_4 = \{\text{identity}, 90^\circ \text{ rotation}, 180^\circ \text{ rotation}, 270^\circ \text{ rotation},\\ \text{horizontal flip}, \text{vertical flip}, \text{transpose}, \text{anti-transpose}\}.$$

For non-square images, TreemapMix uses the shape-preserving subset

$$\{\text{identity}, 180^\circ \text{ rotation}, \text{horizontal flip}, \text{vertical flip}\}.$$

These transformations preserve region areas, so they increase spatial diversity without changing the TreemapMix target.

A.5 Cyclic views and slot exposure

Cyclic views are a tractable alternative to enumerating all $K!$ image-to-slot assignments. If an epoch contains G complete TreemapMix groups, full permutation enumeration would produce $K!G$ mixed images, increasing the sample count by $\frac{K!G}{KG} = (K-1)!$. In contrast, cyclic TreemapMix produces exactly K views per group, giving KG mixed images and preserving the number of training samples after class-balanced row construction.

Cyclic permutation is designed to decouple source labels from fixed spatial slots. Although any individual mixed view may assign a class to a small or large region, the full cyclic orbit exposes every source image to every slot exactly once. Therefore, when the source labels in a group are distinct, every participating class is paired once with every area weight and, for PL training, every rank position induced by the sorted slot weights. Appendix Lemma 3 formalizes the resulting slot-exposure balance.

B Soft TreemapMix Cross-Entropy Derivations

We use the sparse target notation from Section 3.2. For one TreemapMix sample, let $\mathcal{T} = \{(c_i, \eta_i)\}_{i=1}^{K}$, where $\eta_i \geq 0$ and $\sum_{i=1}^{K} \eta_i = 1$, where c_i is the class assigned to slot i and η_i is the corresponding slot weight. For cyclic view s, we have $c_i = y_{\pi_s(i)}$ and $\eta_i = \lambda_i$.

B.1 KL derivation

The sparse target distribution is $q_c = \sum_{i=1}^{K} \eta_i \mathbf{1}[c_i = c]$. It is a valid probability distribution because

$$\sum_{c=1}^{C} q_c = \sum_{c=1}^{C} \sum_{i=1}^{K} \eta_i \mathbf{1}[c_i = c] = \sum_{i=1}^{K} \eta_i \sum_{c=1}^{C} \mathbf{1}[c_i = c] = \sum_{i=1}^{K} \eta_i = 1.$$

Hence $q = (q_1, \ldots, q_C)$ lies on the probability simplex.

The KL divergence from q to p_θ is

$$D_{\mathrm{KL}}(q \| p_\theta) = \sum_{c=1}^{C} q_c \log \frac{q_c}{p_\theta(c \mid \tilde{x})} = \sum_{c=1}^{C} q_c \log q_c - \sum_{c=1}^{C} q_c \log p_\theta(c \mid \tilde{x}).$$

The first term is independent of θ. Thus, minimizing the KL divergence is equivalent to minimizing soft cross-entropy. Substituting the sparse target gives

$$\begin{aligned} \mathcal{L}_{\text{TM-CE}} &= -\sum_{c=1}^{C} \left(\sum_{i=1}^{K} \eta_i \mathbf{1}[c_i = c] \right) \log p_\theta(c \mid \tilde{x}) \\ &= -\sum_{i=1}^{K} \eta_i \log p_\theta(c_i \mid \tilde{x}), \end{aligned}$$

which is the sparse form in (5). For a mini-batch of size B,

$$\mathcal{L}_{\text{TM-CE}} = -\frac{1}{B} \sum_{b=1}^{B} \sum_{i=1}^{K} \eta_{b,i} \log p_\theta(c_{b,i} \mid \tilde{x}_b).$$

B.2 Gradient derivation

For one TreemapMix sample, write

$$\mathcal{L}_{\text{TM-CE}} = -\sum_{c=1}^{C} q_c \log p_c, \qquad p_c = \frac{\exp(z_c)}{\sum_{j=1}^{C} \exp(z_j)}.$$

Since $\sum_{c=1}^{C} q_c = 1$, the log-softmax identity gives

$$\begin{aligned} \log p_c &= z_c - \log \sum_{j=1}^{C} \exp(z_j), \\ \mathcal{L}_{\text{TM-CE}} &= -\sum_{c=1}^{C} q_c z_c + \log \sum_{j=1}^{C} \exp(z_j), \\ \frac{\partial \mathcal{L}_{\text{TM-CE}}}{\partial z_i} &= -q_i + \frac{\exp(z_i)}{\sum_{j=1}^{C} \exp(z_j)} = p_i - q_i. \end{aligned}$$

Substituting $q_i = \sum_j \eta_j \mathbf{1}[c_j = i]$ gives the sparse gradient. For a mini-batch of size B,

$$\frac{\partial \mathcal{L}_{\text{TM-CE}}}{\partial z_{b,c}} = \frac{1}{B} \left[p_\theta(c \mid \tilde{x}_b) - \sum_{i=1}^{K} \eta_{b,i} \mathbf{1}[c_{b,i} = c] \right].$$

B.3 Random cyclic-view target variance

Lemma 1 (Random cyclic-view and cyclic-average target variance). Assume distinct source labels, an unfiltered draw $w \sim \mathrm{Dirichlet}(\alpha, \ldots, \alpha)$ with $\alpha > 0$, and let $q_c^{(s)} = \sum_{i=1}^{K} \lambda_i \mathbf{1}[y_{\pi_s(i)} = c]$ be the TreemapMix-SCE target mass for class c in cyclic view s. Conditioned on the source labels, if c is a participating class and s is sampled uniformly from $\{0, \ldots, K-1\}$ independently of w, then

$$\mathbb{E}_{w,s}[q_c^{(s)}] = \frac{1}{K}, \qquad \mathrm{Var}_{w,s}(q_c^{(s)}) = \frac{K-1}{K^2(K\alpha+1)}.$$

Moreover, averaging over the full cyclic orbit gives $\frac{1}{K}\sum_{s=0}^{K-1} q_c^{(s)} = \frac{1}{K}$, so the cyclic-average target mass has zero variance conditioned on the source labels. For an absent class, $q_c^{(s)} = 0$ for all s.

Proof. Random cyclic view. Assume the K source labels are distinct, and let $c = y_k$ be a participating class. For each cyclic view s, class c appears in exactly one slot. Since the cyclic map visits every slot exactly once as s ranges from 0 to $K-1$, sampling s uniformly makes the slot index containing class c uniformly distributed over $\{1, \ldots, K\}$. Therefore, $q_c^{(s)} = \lambda_I$, where $I \sim \mathrm{Uniform}\{1, \ldots, K\}$. Because $(\lambda_1, \ldots, \lambda_K)$ is a sorted permutation of $(w_1, \ldots, w_K)$, a uniformly selected sorted component λ_I has the same marginal distribution as a uniformly selected unsorted component w_J. By exchangeability of the symmetric Dirichlet distribution, this marginal distribution is the same as that of any component w_j. Hence $\mathbb{E}_{w,s}[q_c^{(s)}] = \mathbb{E}[w_j] = \frac{1}{K}$. For a symmetric Dirichlet distribution with concentration parameter α in each coordinate, $\mathrm{Var}(w_j) = \frac{K-1}{K^2(K\alpha+1)}$. Therefore, $\mathrm{Var}_{w,s}(q_c^{(s)}) = \mathrm{Var}(w_j) = \frac{K-1}{K^2(K\alpha+1)}$.

Full cyclic orbit. The cyclic-average identity follows from Lemma 3. When labels are distinct, $m_c = 1$ for every participating class, so $\frac{1}{K}\sum_{s=0}^{K-1} q_c^{(s)} = \frac{1}{K}$. Since this equality holds for every Dirichlet sample, the conditional variance of the cyclic-average target mass is zero. If c is absent from the group, then no slot carries class c, and $q_c^{(s)} = 0$ for all s.

Fixed-logit interpretation. If logits are fixed, then $\frac{\partial \mathcal{L}_{\text{TM-CE}}}{\partial z_c} = p_c - q_c^{(s)}$ has the same variance as $q_c^{(s)}$, since p_c is held fixed while w and s vary. This is a fixed-logit target-side calculation, not a formula for the stochastic network-gradient variance when logits depend on the mixed input. Layout rejection and uniform fallback alter the weight distribution, so the exact Dirichlet variance need not hold for the filtered training sampler. The full-orbit identity holds for any normalized weights reused across all views.

□

C TreemapMix Plackett–Luce Derivations

The derivation assumes K distinct target classes. Let $r_1 \succ \cdots \succ r_K$ denote their descending order by area weight, set $L = K$, and write ρ_{r_ℓ} for the area weight assigned to class r_ℓ. Let $\mathcal{U} = \{1, \ldots, C\} \setminus \{r_1, \ldots, r_K\}$ be the absent classes, with total predicted probability mass $p_{\mathcal{U}} = 1 - \sum_{j=1}^{K} p_{r_j}$.

C.1 From the Plackett–Luce probability to TreemapMix-PL

Let the model output logits $z = f_\theta(\tilde{x}) \in \mathbb{R}^C$ and softmax probabilities $p_c = p_\theta(c \mid \tilde{x})$. For the descending ranking $r_1 \succ \cdots \succ r_L$, the top-$L$ Plackett–Luce probability with unranked residual mass is

$$P_{\mathrm{PL}}(r_1 \succ \cdots \succ r_L \mid \tilde{x}) = \prod_{\ell=1}^{L} \frac{p_{r_\ell}}{p_{\mathcal{U}} + \sum_{j=\ell}^{L} p_{r_j}}.$$

The corresponding unweighted ListMLE-style negative log-likelihood is

$$\mathcal{L}_{\mathrm{ListMLE}} = -\sum_{\ell=1}^{L} \log \frac{p_{r_\ell}}{p_{\mathcal{U}} + \sum_{j=\ell}^{L} p_{r_j}}.$$

TreemapMix-PL weights each ranking term by its area weight ρ_{r_ℓ}:

$$\mathcal{L}_{\text{TM-PL}} = -\sum_{\ell=1}^{L} \rho_{r_\ell} \log \frac{p_{r_\ell}}{p_{\mathcal{U}} + \sum_{j=\ell}^{L} p_{r_j}}.$$

This is the descending form in (9). This weighted stage-loss surrogate is not generally a normalized likelihood over rankings. With equal weights it is a constant rescaling of the unweighted top-K ListMLE loss. Equal-weight entries retain their source-slot order, including under uniform fallback.

C.2 Ascending implementation form

For implementation, we reverse the descending order. Define $a_i = r_{L+1-i}$, $\gamma_i = \rho_{a_i}$, and $\gamma_1 \le \gamma_2 \le \cdots \le \gamma_L$. Thus, a_1 is the weakest ranked class and a_L is the strongest. Reindexing the descending objective gives

$$\mathcal{L}_{\text{TM-PL}} = -\sum_{i=1}^{L} \gamma_i \log \frac{p_{a_i}}{p_{\mathcal{U}} + \sum_{j=1}^{i} p_{a_j}}.$$

Using

$$D_i = p_{\mathcal{U}} + \sum_{j=1}^{i} p_{a_j} = 1 - \sum_{j=i+1}^{L} p_{a_j},$$

we obtain

$$\mathcal{L}_{\text{TM-PL}} = \sum_{i=1}^{L} \gamma_i \left[\log D_i - \log p_{a_i}\right],$$

which is the ascending implementation form in (11). For a mini-batch of size B, where sample b has L_b distinct ranked classes, the corresponding batch objective is

$$\mathcal{L}_{\text{TM-PL}} = \frac{1}{B} \sum_{b=1}^{B} \sum_{i=1}^{L_b} \gamma_{b,i} \left[\log D_{b,i} - \log p_\theta(a_{b,i} \mid \tilde{x}_b)\right].$$

C.3 Gradient derivation

We differentiate the ascending objective above, holding its target weights and rank assignments fixed. Define the stronger-class probability mass $R_i = \sum_{j=i+1}^{L} p_{a_j}$ and $D_i = 1 - R_i$, and the stronger-class set $T_i = \{a_{i+1}, \ldots, a_L\}$. First, $\frac{\partial \log p_{a_i}}{\partial z_c} = \mathbf{1}[c = a_i] - p_c$. Next,

$$\begin{aligned}\frac{\partial R_i}{\partial z_c} &= \sum_{j=i+1}^{L} p_{a_j} \left(\mathbf{1}[c = a_j] - p_c\right) \\ &= p_c \mathbf{1}[c \in T_i] - p_c R_i \\ &= p_c \left(\mathbf{1}[c \in T_i] - R_i\right).\end{aligned}$$

The denominator derivatives are therefore

$$\begin{aligned}\frac{\partial D_i}{\partial z_c} &= -\frac{\partial R_i}{\partial z_c} = p_c \left(R_i - \mathbf{1}[c \in T_i]\right), \\ \frac{\partial \log D_i}{\partial z_c} &= \frac{p_c \left(R_i - \mathbf{1}[c \in T_i]\right)}{D_i}.\end{aligned}$$

For the i-th term $\ell_i = \gamma_i[\log D_i - \log p_{a_i}]$, we have

$$\begin{aligned}\frac{\partial \ell_i}{\partial z_c} &= \gamma_i \left[\frac{p_c \left(R_i - \mathbf{1}[c \in T_i]\right)}{D_i} - \mathbf{1}[c = a_i] + p_c\right] \\ &= \gamma_i \left[\frac{p_c \left(R_i - \mathbf{1}[c \in T_i] + D_i\right)}{D_i} - \mathbf{1}[c = a_i]\right] \\ &= \gamma_i \left[\frac{p_c \left(1 - \mathbf{1}[c \in T_i]\right)}{D_i} - \mathbf{1}[c = a_i]\right],\end{aligned}$$

because $R_i + D_i = 1$. Summing over i gives

$$\frac{\partial \mathcal{L}_{\text{TM-PL}}}{\partial z_c} = \sum_{i=1}^{L} \gamma_i \left[\frac{p_c \left(1 - \mathbf{1}[c \in T_i]\right)}{D_i} - \mathbf{1}[c = a_i]\right],$$

which is the gradient in (13).

C.4 Ordinal gradient structure

The gradient above has a simple ordinal interpretation.

Unranked class

Let $u \notin \{a_1, \ldots, a_L\}$. Then $\mathbf{1}[u \in T_i] = 0$ and $\mathbf{1}[u = a_i] = 0$ for all i, so $\frac{\partial \mathcal{L}_{\text{TM-PL}}}{\partial z_u} = p_u \sum_{i=1}^{L} \frac{\gamma_i}{D_i} \geq 0$. Thus, direct gradient descent on logits suppresses absent classes. Updates through shared network parameters need not move each individual logit in that direction.

Ranked class

Let $c = a_m$ for some $m \in \{1, \ldots, L\}$. For $i < m$, class a_m belongs to the stronger set T_i, so those terms vanish. For $i = m$, the contribution is $\gamma_m \left(\frac{p_{a_m}}{D_m} - 1\right)$. For $i > m$, class a_m is not in T_i and is not equal to a_i, so the remaining contribution is $p_{a_m} \sum_{i=m+1}^{L} \frac{\gamma_i}{D_i}$. Therefore,

$$\frac{\partial \mathcal{L}_{\text{TM-PL}}}{\partial z_{a_m}} = \gamma_m \left(\frac{p_{a_m}}{D_m} - 1\right) + p_{a_m} \sum_{i=m+1}^{L} \frac{\gamma_i}{D_i}.$$

For the strongest class a_L, the sum is empty and $D_L = 1$, so $\frac{\partial \mathcal{L}_{\text{TM-PL}}}{\partial z_{a_L}} = \gamma_L (p_{a_L} - 1) \leq 0$. Thus, direct logit gradient descent promotes the strongest ranked class unless $p_{a_L} = 1$, suppresses unranked classes, and regulates intermediate ranked classes through competition with stronger classes.

C.5 Proof of the Dirichlet order-statistics result

Lemma 2 (Dirichlet order statistics and ranking emphasis). Assume distinct target labels and unfiltered symmetric Dirichlet weights with $\alpha > 0$, so that $L = K$ and $\gamma_i = \lambda_i$. Let $\mu_i(\alpha) = \mathbb{E}[\lambda_i]$ denote the expected i-th sorted Dirichlet weight. Holding logits and rank assignments fixed while varying the weights, the expected TreemapMix-PL gradients satisfy

$$\mathbb{E}_w\left[\frac{\partial \mathcal{L}_{\text{TM-PL}}}{\partial z_{a_m}}\right] = \mu_m(\alpha)\left(\frac{p_{a_m}}{D_m} - 1\right) + p_{a_m}\sum_{i=m+1}^{K}\frac{\mu_i(\alpha)}{D_i},$$

$$\mathbb{E}_w\left[\frac{\partial \mathcal{L}_{\text{TM-PL}}}{\partial z_u}\right] = p_u\sum_{i=1}^{K}\frac{\mu_i(\alpha)}{D_i}, \qquad u \notin \{a_1, \ldots, a_K\}.$$

Proof. Expected gradient. Assume the K target labels are distinct, so that $L = K$ and the ascending PL weights satisfy $\gamma_i = \lambda_i$. Let $\mu_i(\alpha) = \mathbb{E}[\lambda_i]$ denote the expected i-th sorted Dirichlet weight. Holding logits and rank assignments fixed while varying the weights, D_i and p_c are fixed, and the ranked-class gradient is linear in λ_i. Taking expectations term by term replaces each λ_i by $\mu_i(\alpha)$, giving both identities in the lemma.

The expected fixed-logit gradient therefore depends on α through $\mu_i(\alpha)$. This linearity argument does not establish monotonicity of individual gaps between order statistics.

Large-concentration limit. For $\alpha \to \infty$, each coordinate has mean $1/K$ and variance $(K - 1)/(K^2(K\alpha + 1)) \to 0$. A union bound implies $\max_j |w_j - 1/K| \to 0$ in probability, and sorting preserves this convergence. Thus the loss tends to $1/K$ times the unweighted top-K objective for fixed logits and rank assignments.

Scope. For filtered sampling the same linearity uses moments of the actual accepted/fallback distribution, rather than unfiltered Dirichlet moments. These calculations hold predictions fixed; they do not describe the full training-gradient expectation when predictions vary with the mixed input.

□

D Proof of Cyclic Permutation Balancing

Lemma 3 (Cyclic permutation balances slot and target exposure) Let the normalized slot weights satisfy $\sum_{i=1}^{K} \lambda_i = 1$, and define

$$q_c^{(s)} = \sum_{i=1}^{K} \lambda_i \mathbf{1}[y_{\pi_s(i)} = c], \qquad m_c = \sum_{k=1}^{K} \mathbf{1}[y_k = c]. \tag{16}$$

Then, for every class c and slot i,

$$\frac{1}{K} \sum_{s=0}^{K-1} \mathbf{1}[y_{\pi_s(i)} = c] = \frac{m_c}{K}, \tag{17}$$

and consequently

$$\frac{1}{K} \sum_{s=0}^{K-1} q_c^{(s)} = \frac{m_c}{K}. \tag{18}$$

Proof. Slot exposure. For cyclic view s, slot i receives source image $x_{\pi_s(i)}$ and label $y_{\pi_s(i)}$, where $\pi_s(i) = 1 + \big((i + s - 1) \bmod K\big)$. For any fixed slot i, as s ranges from 0 to $K - 1$, the index $\pi_s(i)$ visits every source position exactly once. Therefore, for any class c, $\sum_{s=0}^{K-1} \mathbf{1}[y_{\pi_s(i)} = c] = \sum_{k=1}^{K} \mathbf{1}[y_k = c] = m_c$. Dividing by K gives the slot-balancing identity $\frac{1}{K} \sum_{s=0}^{K-1} \mathbf{1}[y_{\pi_s(i)} = c] = \frac{m_c}{K}$, which proves (17).

Target exposure. The soft target mass assigned to class c in cyclic view s is $q_c^{(s)} = \sum_{i=1}^{K} \lambda_i \mathbf{1}[y_{\pi_s(i)} = c]$. Averaging over cyclic views gives

$$\begin{aligned}
\frac{1}{K} \sum_{s=0}^{K-1} q_c^{(s)} &= \frac{1}{K} \sum_{s=0}^{K-1} \sum_{i=1}^{K} \lambda_i \mathbf{1}[y_{\pi_s(i)} = c] \\
&= \frac{1}{K} \sum_{i=1}^{K} \lambda_i \sum_{s=0}^{K-1} \mathbf{1}[y_{\pi_s(i)} = c] \\
&= \frac{1}{K} \sum_{i=1}^{K} \lambda_i m_c \\
&= \frac{m_c}{K},
\end{aligned}$$

because $\sum_{i=1}^{K} \lambda_i = 1$. This proves (18).

When the K source labels are distinct, $m_c = 1$ for every participating class and $m_c = 0$ for every absent class. Therefore, each participating class appears exactly once in every slot across the K cyclic views. Since slots are associated with sorted weights and rank positions, cyclic permutation also prevents any participating class from being permanently tied to a fixed TreemapMix weight or rank. The PL denominator terms remain prediction-dependent, so this balancing does not remove ordinal competition; it only removes persistent source-label-to-slot bias.

□

E Implementation Details

E.1 Experimental Protocol and Hyperparameters

Tables 10, 11, and 12 summarize the ImageNet-1K experimental protocol and baseline hyperparameters. Table 10 summarizes the schedule-level adaptations applied to the reference `timm` recipes for the main 110-epoch comparison. Tables 11 and 12 separate data-augmentation settings from optimizer and training settings for readability.

The baseline recipes are adapted from established `timm` ImageNet-1K training configurations [43, 44]. Because our main study uses a shorter 110-epoch budget than many reference recipes, we adapt the training recipe: we disable `aug_repeats`, adjust the learning rate, disable gradient clipping, and keep the remaining regularization settings fixed across methods. All methods therefore share the same architecture, data budget, optimizer schedule, and evaluation protocol. The comparison isolates augmentation and loss effects, but does not claim per-baseline optimality.

Runs were executed on H20 nodes with eight GPUs and 2 TB system RAM, with independent two-GPU jobs colocated by a custom scheduler. Across reported main and appendix experiments, training used approximately 5.5k GPU-hours, while peak observed host-memory use remained below 400 GB per node.

The main ImageNet-1K comparison uses a global batch size of 1024. We omit epochs and warmup epochs from Tables 11 and 12 because those values are held fixed across runs. The augmentation table lists the ImageNet-1K augmentation recipes used in the main comparisons.

For the main comparison, RICAP uses the following auxiliary settings: the augmentation is applied with probability 0.5, disabled for the final 10 epochs, allowed to vary the crop center over a ratio range from 0.5 to 1.5, and allowed to vary the crop scale from 0.8 to 1.2. In this comparison, TreemapMix-SCE and TreemapMix-mixed runs use $K = 4$ with a cosine α schedule from 0.1 to 0.5, while TreemapMix-PL runs use $K = 6$ with the same cosine α schedule.

E.2 Compositional diagnostic protocol

We evaluate the ViT-Betwixt checkpoints from the main ImageNet-1K experiments reported in Figure 2 across three seeds on a held-out composed diagnostic set. The set contains 500 TreemapMix samples for each $K \in \{3, 4, 5, 6\}$. Figure 3 reports three diagnostics, all restricted to classes present in each mixture: pairwise rank accuracy is the fraction of class pairs whose logits follow their area ordering (left), counting only pairs with area fractions differing by at least 0.05; Spearman correlation measures agreement between area and logit ranks (center); and largest-area top-class accuracy is the fraction of samples whose largest-area class has the highest logit among present classes (right). Spearman excludes samples with tied areas or logits; top-class accuracy excludes tied largest areas.

E.3 COCO transfer setup

Table 13 summarizes the ViTDet-style configurations used for the COCO transfer experiments. In all cases the starting point is set to the corresponding checkpoint from the main classification experiments, and we fine-tune a Mask R-CNN model with a ViTDet-style adapter for 30 epochs on COCO.

We compare representations learned by different ImageNet-1K pretraining augmentations under a fixed COCO fine-tuning setup. We evaluate ViT-Wee and ViT-Betwixt to cover smaller and larger backbones. Following the ViTDet setup [27], we use a ViTDet-style adapter and train a Mask R-CNN detector [15, 27]. We fine-tune on COCO for 30 epochs, initializing from the checkpoints produced in the main classification experiments. COCO images are resized to 256×256 pixels to match the input resolution used during ImageNet-1K pretraining.

We evaluate TreemapMix-SCE and TreemapMix-PL and compare them with Mixup+CutMix-pretrained and RICAP-pretrained baselines [39]. We include RICAP because it is the closest multi-image composition baseline in our study and is related to Mosaic-style object-detection augmentations used in YOLOv4 and YOLOv5 [3, 20]. The corresponding fine-tuning hyperparameters are reported in Appendix E.3. Table 9 reports COCO box and mask average precision (AP), averaged

Table 10: Adaptations to the reference `timm` ImageNet-1K recipes used for the main 110-epoch comparison. We shorten the original long schedules, disable augmentation repeats, switch to `bfloat16`, and retune the learning rate; all other recipe settings are kept unchanged unless listed.

Model	Setting	Original	New
ViT-Betwixt			
	`dtype`	`float16`	`bfloat16`
	`aug_repeats`	3.0	0.0
	`warmup_epochs`	20	10
	`epochs`	600	110
	`lr`	1.2e-3	1.5e-3
ViT-Medium			
	`dtype`	`float16`	`bfloat16`
	`aug_repeats`	3.0	0.0
	`clip_grad`	5.0	`null`
	`warmup_epochs`	20	10
	`epochs`	600	110
	`lr`	1.2e-3	1.5e-3
ViT-Little			
	`dtype`	`float16`	`bfloat16`
	`aug_repeats`	3.0	0.0
	`clip_grad`	5.0	`null`
	`warmup_epochs`	20	10
	`epochs`	600	110
	`lr`	1.2e-3	1.5e-3
ViT-Wee			
	`dtype`	`float16`	`bfloat16`
	`aug_repeats`	3.0	0.0
	`clip_grad`	5.0	`null`
	`warmup_epochs`	20	10
	`epochs`	450	110
	`lr`	1.2e-3	1.5e-3

Table 11: ViT data-augmentation hyperparameters for the ImageNet-1K main comparison recipes. A checkmark in the Single-Image Aug. column indicates that the recipe includes standard image-only transforms such as cropping, flipping, scaling, color perturbations, AutoAugment-style transformations [8], and random erasing. The fixed auxiliary RICAP and TreemapMix settings are described in the appendix text.

Config	Single-Image Aug	Mixup	CutMix	Mixup Prob	CutMix Prob	RICAP Prob
No Aug	×	0.0	0.0	0.0	0.0	0.0
Single-Image Aug	✓	0.0	0.0	0.0	0.0	0.0
Mixup	✓	0.8	0.0	1.0	0.0	0.0
CutMix	✓	0.0	1.0	0.0	1.0	0.0
Mixup+CutMix	✓	0.8	1.0	0.5	0.5	0.0
RICAP	✓	0.0	0.0	0.0	0.0	0.5
TreemapMix-SCE	✓	0.0	0.0	0.0	0.0	0.0
TreemapMix-PL	✓	0.0	0.0	0.0	0.0	0.0
TreemapMix-mixed	✓	0.0	0.0	0.0	0.0	0.0

Table 12: ViT baseline optimizer and training hyperparameters for the main comparison. Opt. denotes optimizer, LR learning rate, WD weight decay, and EMA exponential moving average of model weights. NAdamW is Nesterov-accelerated Adam with decoupled weight decay. DropPath is the stochastic-depth probability.

Model	Opt.	Base LR	LR Decay	WD	DropPath	EMA
ViT-Wee	NAdamW	1.5e-3	cosine	0.06	0.2	yes
ViT-Little	NAdamW	1.5e-3	cosine	0.06	0.2	no
ViT-Medium	NAdamW	1.5e-3	cosine	0.06	0.2	no
ViT-Betwixt	NAdamW	1.5e-3	cosine	0.08	0.2	no

over three transferred checkpoints whose ImageNet-1K pretraining used different random seeds. Appendix Table 14 retains the corresponding box and mask AP50 results.

Table 13: Hyperparameters for the object-detection transfer experiments with a ViTDet-style adapter and Mask R-CNN on COCO. Embed Dim is the embedding dimension, Heads is the attention-head count, MLP Ratio is the multilayer perceptron expansion ratio, and LR is the learning rate.

Model	Embed Dim	Depth	Heads	MLP Ratio	Drop Path	LR	Layer Decay
ViT-Wee	256	14	4	5.0	0.1	6.0e-4	0.7
ViT-Betwixt	640	12	10	4.0	0.1	8.0e-4	0.7

The corresponding box and mask AP50 results are retained in Table 14. They show the same ordering as the COCO AP comparison, with PL highest on both backbones.

Table 14: COCO transfer AP50 after ViTDet-style Mask R-CNN fine-tuning from ImageNet-1K checkpoints. In this transfer setting, TreemapMix-PL obtains the best box and mask AP50 for both ViT-Wee and ViT-Betwixt, while TreemapMix-SCE also improves over Mixup+CutMix and RICAP. Values are mean $\pm$ standard deviation over three seeds; higher is better, and bold marks the best mean in each column.

	ViT-Wee		ViT-Betwixt	
Method	Box AP50 $\uparrow$	Mask AP50 $\uparrow$	Box AP50 $\uparrow$	Mask AP50 $\uparrow$
Mixup+CutMix	39.67 ± 0.01	35.64 ± 0.09	46.46 ± 0.19	42.59 ± 0.30
RICAP	38.68 ± 0.24	34.82 ± 0.20	44.91 ± 0.04	41.05 ± 0.20
TreemapMix-SCE	40.37 ± 0.19	36.18 ± 0.19	47.06 ± 0.15	43.12 ± 0.13
TreemapMix-PL	$\mathbf{40.90 \pm 0.01}$	$\mathbf{36.59 \pm 0.13}$	$\mathbf{47.58 \pm 0.27}$	$\mathbf{43.48 \pm 0.14}$

E.4 Loss functions in code

We include concise PyTorch reference implementations of TreemapMix-SCE (Figure 4), TreemapMix-PL (Figure 5), and the mixed-loss variant (Figure 6) for the distinct-label mathematical objectives. The PL listing is a numerically stable reference, not a reconstruction of every branch of the experimental training code.

```python
import torch.nn.functional as F

def treemapmix_sce_loss(logits, labels, weights):
    """
    logits: model logits, shape (B, C)
    labels: target class indices, shape (B, K)
    weights: target area weights, shape (B, K)
    Each pair (labels[b, i], weights[b, i]) represents one
    TreemapMix region for sample b.
    """
    log_probs = F.log_softmax(logits, dim=1)
    target_log_probs = log_probs.gather(dim=1, index=labels)
    losses = -(weights * target_log_probs).sum(dim=1)
    return losses.mean()
```

Figure 4: TreemapMix-SCE. The loss matches the area-weighted soft target induced by the TreemapMix regions.

```python
import torch

def treemapmix_pl_loss(logits, labels, weights):
    # logits: (B, C); labels and weights: (B, K)
    # Distinct labels; ascending area order; weights sum to 1.
    ranked = logits.gather(1, labels)
    absent = torch.ones_like(logits, dtype=torch.bool)
    absent.scatter_(1, labels, False)
    absent_lse = torch.logsumexp(
        logits.masked_fill(~absent, -torch.inf), dim=1)
    prefix_lse = torch.logcumsumexp(ranked, dim=1)
    denom_lse = torch.logaddexp(
        absent_lse[:, None], prefix_lse)
    return (weights * (denom_lse - ranked)).sum(1).mean()
```

Figure 5: Numerically stable reference for the distinct-label TreemapMix-PL objective, used by the new TreemapMix package. Log-domain remaining-set normalization avoids subtracting probabilities from one. The historical LabelMix training implementation uses probability-space clamping and a separate repeated-entry surrogate.

```python
def treemapmix_mixed_loss(logits, labels, weights, beta=0.5):
    sce = treemapmix_sce_loss(logits, labels, weights)
    pl = treemapmix_pl_loss(logits, labels, weights)
    return beta * sce + (1.0 - beta) * pl
```

Figure 6: TreemapMix-mixed, using the functions in Figures 4 and 5. The coefficient β matches the notation in the main text.

E.5 Runtime overhead

For the configurations in this benchmark, augmentations are applied in the data loader. Table 7 isolates data-loader throughput and host-memory usage for six augmentation configurations. The benchmark measures CPU-side loading and augmentation without model forward or backward passes. Peak resident set size (RSS) and unique set size (USS) are summed over the loader process tree. RSS can count shared pages more than once, while USS measures private memory. Memory is reported in GiB.

Relative to the single-image recipe (1282.8 images/s), TreemapMix-SCE at $K = 4$ reaches 1039.4 images/s, a 19.0% throughput reduction; TreemapMix-PL at $K = 6$ reaches 954.3 images/s, a 25.6% reduction. The corresponding peak USS increases from 17.76 GiB to 20.16 and 21.74 GiB. Mixup+CutMix reaches 1223.4 images/s, a 4.6% reduction. The RICAP configuration reaches 364.8 images/s with 25.18 GiB peak USS. TreemapMix therefore adds measurable host-side cost.

The training configuration uses a prefetch factor of 2, which allows CPU preparation to overlap GPU computation. The isolated loader benchmark does not measure how much of this overhead is hidden during training, so it does not establish an end-to-end per-epoch timing comparison.

F Additional Experiments

F.1 Full ImageNet-1K Results

Tables 1 and 2 report the full ImageNet-1K main-comparison Top-1 accuracy and ECE results in the main text, respectively. Figure 2 summarizes the same results graphically.

The original comparison also reports negative log-likelihood (NLL) and Brier score in Tables 15 and 16. SCE has the lowest mean NLL on all four backbones; Brier differences are small and sometimes favor the mixed objective. These complementary metrics preserve the broader confidence-quality evaluation beyond ECE.

Table 15: ImageNet-1K negative log-likelihood for the original nine-configuration comparison under the matched 110-epoch protocol. Values are mean ± one sample standard deviation over three seeds. Bold marks the lowest mean in each backbone.

Method	ViT-Wee ↓	ViT-Little ↓	ViT-Medium ↓	ViT-Betwixt ↓
Single-Image Aug	0.923 ± 0.006	0.869 ± 0.008	0.864 ± 0.001	0.855 ± 0.001
No Aug	1.120 ± 0.009	1.137 ± 0.005	1.154 ± 0.006	1.134 ± 0.008
Mixup+CutMix	1.015 ± 0.010	0.896 ± 0.002	0.854 ± 0.008	0.811 ± 0.004
Mixup	1.012 ± 0.004	0.904 ± 0.002	0.862 ± 0.002	0.820 ± 0.002
CutMix	0.994 ± 0.011	0.870 ± 0.004	0.840 ± 0.005	0.787 ± 0.005
RICAP	0.893 ± 0.004	0.846 ± 0.007	0.870 ± 0.008	0.857 ± 0.003
TreemapMix-mixed	0.862 ± 0.008	0.778 ± 0.011	0.739 ± 0.005	0.728 ± 0.003
TreemapMix-PL	0.881 ± 0.012	0.803 ± 0.001	0.773 ± 0.002	0.766 ± 0.008
TreemapMix-SCE	$\mathbf{0.857 \pm 0.003}$	$\mathbf{0.775 \pm 0.002}$	$\mathbf{0.735 \pm 0.004}$	$\mathbf{0.714 \pm 0.007}$

Table 16: ImageNet-1K Brier score for the original nine-configuration comparison under the matched 110-epoch protocol. Values are mean ± one sample standard deviation over three seeds. Bold marks the lowest mean in each backbone.

Method	ViT-Wee ↓	ViT-Little ↓	ViT-Medium ↓	ViT-Betwixt ↓
Single-Image Aug	0.314 ± 0.002	0.294 ± 0.003	0.291 ± 0.001	0.285 ± 0.001
No Aug	0.356 ± 0.002	0.349 ± 0.002	0.344 ± 0.001	0.335 ± 0.001
Mixup+CutMix	0.342 ± 0.002	0.305 ± 0.001	0.292 ± 0.003	0.277 ± 0.002
Mixup	0.340 ± 0.002	0.308 ± 0.001	0.294 ± 0.001	0.280 ± 0.001
CutMix	0.338 ± 0.003	0.298 ± 0.002	0.287 ± 0.001	0.270 ± 0.001
RICAP	0.319 ± 0.002	0.303 ± 0.002	0.304 ± 0.002	0.299 ± 0.001
TreemapMix-mixed	$\mathbf{0.313 \pm 0.002}$	$\mathbf{0.288 \pm 0.003}$	$\mathbf{0.275 \pm 0.001}$	0.269 ± 0.000
TreemapMix-PL	$\mathbf{0.313 \pm 0.003}$	$\mathbf{0.288 \pm 0.001}$	0.277 ± 0.001	0.271 ± 0.002
TreemapMix-SCE	0.314 ± 0.000	0.289 ± 0.001	0.276 ± 0.001	$\mathbf{0.268 \pm 0.002}$

F.2 Long-Tailed Recognition

We evaluate ImageNet-LT [40] using the same four model scales and compare Mixup+CutMix, RICAP, TreemapMix-SCE, and TreemapMix-PL. For TreemapMix, we use a natural-mixed scheduler that preserves every original sample and retains the natural long-tailed class frequencies. The comparisons use the matched 110-epoch protocol and three seeds. Samples are independently shuffled within each class, distributed across the epoch using midpoint-stratified positions, merged, locally shuffled, and partitioned into K-source groups. Within every complete group, cyclic permutation preserves the source-to-slot balance of Lemma 3.

TreemapMix-PL achieves the highest Top-1 accuracy on all four backbones, improving over Mixup+CutMix by 1.40, 1.02, 1.31, and 1.82 percentage points for ViT-Wee, Little, Medium, and Betwixt, respectively. Calibration behaves differently from balanced ImageNet-1K: Mixup+CutMix has the lowest ECE on every backbone, and neither TreemapMix objective retains that advantage. TreemapMix-SCE also falls below the baseline in accuracy. Thus, the accuracy benefit of PL extends to this imbalanced setting, while probability calibration remains sensitive to the class distribution.

F.3 Cross-Architecture Results

To test whether the ImageNet-1K results extend beyond the four primary ViT/EVA02 variants, we evaluate ConvNeXtV2-Nano, DeiT-Tiny, and Swin-Tiny under the matched 110-epoch protocol. We compare Mixup+CutMix, RICAP, TreemapMix-SCE, and TreemapMix-PL, retaining the TreemapMix settings selected for the primary experiments. Table 5 reports accuracy and ECE.

PL gives the highest mean accuracy for each architecture, improving over Mixup+CutMix by 0.66, 2.95, and 0.95 percentage points for ConvNeXtV2-Nano, DeiT-Tiny, and Swin-Tiny. SCE has the lowest ECE for all three, at 2.74, 1.32, and 2.52 percentage points, respectively. The PL–SCE accuracy gap is small on DeiT-Tiny and Swin-Tiny relative to the observed seed variation; we report the mean ordering without claiming statistical significance.

F.4 CIFAR-100 Results

We evaluate all 16 augmentation settings on CIFAR-100 with the same four ViT backbones and 256-pixel inputs. The matched schedule comprises 110 total epochs, including 10 warmup epochs. TreemapMix uses the settings from the primary comparison: $K = 4$ for SCE and the mixed objective, $K = 6$ for PL, and a cosine α schedule from 0.1 to 0.5. Figure 7 summarizes Top-1 accuracy and ECE with 15 bins; Tables 3 and 4 give all numerical results.

TreemapMix-SCE has the lowest ECE on all four backbones, ranging from 1.62 to 1.87 percentage points. TreemapMix-PL has the highest Top-1 accuracy on ViT-Wee, Medium, and Betwixt, reaching 75.72%, 75.97%, and 76.39%, respectively. On ViT-Little, ResizeMix reaches 75.54% compared with 75.42% for PL. The mixed objective does not consistently improve on either individual objective. These results retain the accuracy–calibration trade-off, without a uniform accuracy advantage across every model.

F.5 Optimizer Robustness

We next test whether the gains of TreemapMix persist across different optimizers on ImageNet-1K. We compare Muon [21], AdamW (Adam with decoupled weight decay) [30], and NAdamW [9, 22]. We keep the core TreemapMix hyperparameters fixed across optimizers. Because the main experiments use NAdamW with a learning rate of 1.5×10^{-3}, we retain this value for NAdamW and also use it for AdamW to keep the Adam-style optimizer comparison matched. For Muon, we use a separately tuned learning rate of 5×10^{-3}.

Table 17 compares Mixup+CutMix, TreemapMix-SCE, and TreemapMix-PL under AdamW, Muon, and NAdamW for the settings in the caption. These results should be interpreted as an optimizer-sensitivity check rather than a definitive optimizer ranking. Across the tested optimizers, the main qualitative pattern from the primary ImageNet-1K comparison is preserved: TreemapMix-PL generally gives the highest Top-1 accuracy, whereas TreemapMix-SCE gives the lowest ECE in most settings. Muon gives the strongest Top-1 accuracy in several configurations, but this gain is often accompanied by higher ECE. AdamW and NAdamW are more competitive on calibration, although neither dominates across both metrics. Overall, these single-seed results suggest that TreemapMix's accuracy and calibration trends are not specific to one optimizer, while the absolute accuracy-calibration trade-off still depends on the optimizer.

F.6 ImageNet-1K Hyperparameter Sensitivity

TreemapMix introduces two main hyperparameters: the Dirichlet concentration α and the number of mixed regions K. We study their effects with a single-seed hyperparameter sweep. The sweep identifies a stable operating regime for both TreemapMix losses: smaller α values and small-to-

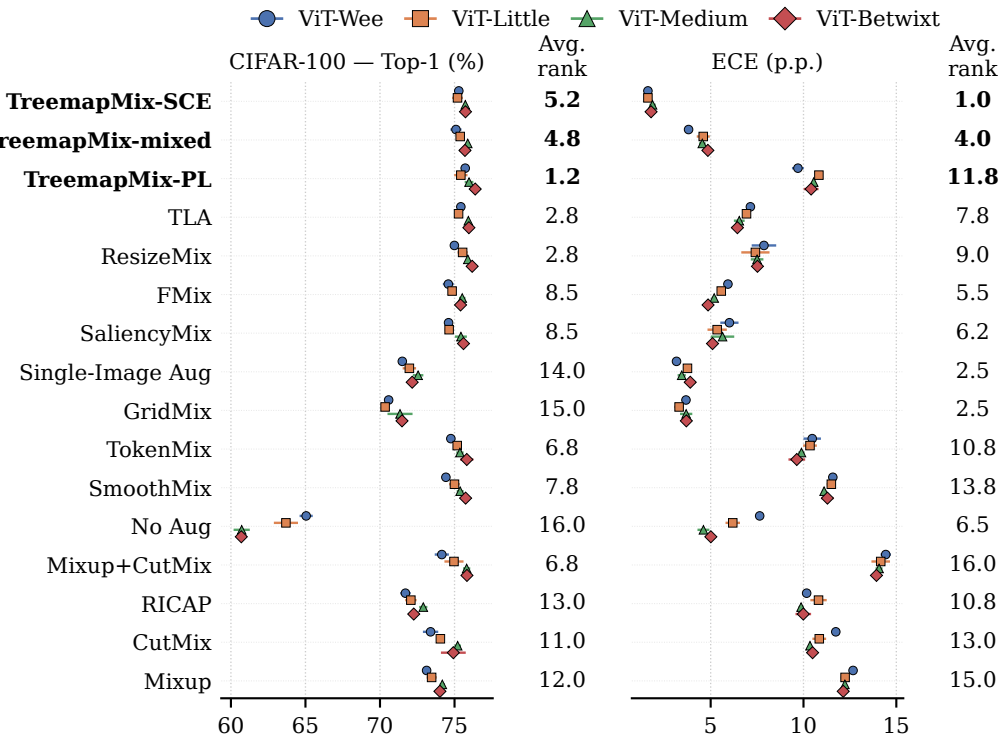


Figure 7: CIFAR-100 comparison at 256-pixel resolution under a 110-epoch schedule, including 10 warmup epochs. Points show the mean over seeds 42, 43, and 44; intervals denote ± one sample standard deviation. ECE uses 15 bins and is reported in percentage points. Average ranks summarize within-backbone ranks, with rank 1 best. Complete values appear in Tables 3 and 4.

Table 17: ImageNet-1K optimizer comparison under the settings in Appendix F.5, using one run with seed 42 per configuration. Accuracy is reported in percent and ECE in percentage points; bold marks the best value for each metric within each row.

		Mixup+CutMix		TreemapMix-SCE		TreemapMix-PL	
Model	Optimizer	Acc. ↑	ECE ↓	Acc. ↑	ECE ↓	Acc. ↑	ECE ↓
ViT-Wee	AdamW	75.76	8.99	77.18	**0.43**	**77.75**	5.41
	Muon	77.92	9.88	78.80	**2.04**	**79.05**	5.93
	NAdamW	76.30	9.29	77.48	**0.94**	**78.16**	5.44
ViT-Little	AdamW	78.77	8.11	79.50	**1.36**	**79.93**	5.96
	Muon	80.27	8.61	80.72	**3.05**	**80.87**	6.57
	NAdamW	78.80	7.65	79.72	**1.90**	**80.09**	6.23
ViT-Medium	AdamW	79.89	8.25	80.33	**0.87**	**80.71**	5.93
	Muon	80.98	7.86	81.42	**2.77**	**81.43**	6.49
	NAdamW	80.04	7.69	80.73	**2.09**	**81.01**	6.39
ViT-Betwixt	AdamW	81.19	8.11	81.00	**2.02**	**81.51**	7.05
	Muon	81.71	7.32	**81.85**	**3.32**	81.79	7.15
	NAdamW	81.13	7.50	81.26	**2.75**	**81.80**	7.18

medium values of K perform best, while more extreme compositions degrade both accuracy and calibration. Overall, TreemapMix is more sensitive to α than to K.

We sweep $\alpha \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0, 3.0\}$ and $K \in \{2, \ldots, 10\}$ using the same training hyperparameters as in the main run. This sweep uses a simplified fixed-hyperparameter recipe: it does not apply aspect-ratio filtering to TreemapMix regions and does not schedule either α or K. The retained plots contain the available $K \in \{3, \ldots, 10\}$ results; the stated sweep also includes $K = 2$, whose points are not present in these assets. No missing points are interpolated. Therefore, these results should be interpreted as a robustness and sensitivity analysis rather than as a best-case configuration search. The absolute numbers are consequently lower than the tuned main results.

Figure 8 shows the resulting sensitivity plots and includes reference results for the main Mixup+CutMix and single-image-only baselines. We include the single-image-only baseline because it is the strongest non-TreemapMix setting in the main comparison for ViT-Wee.

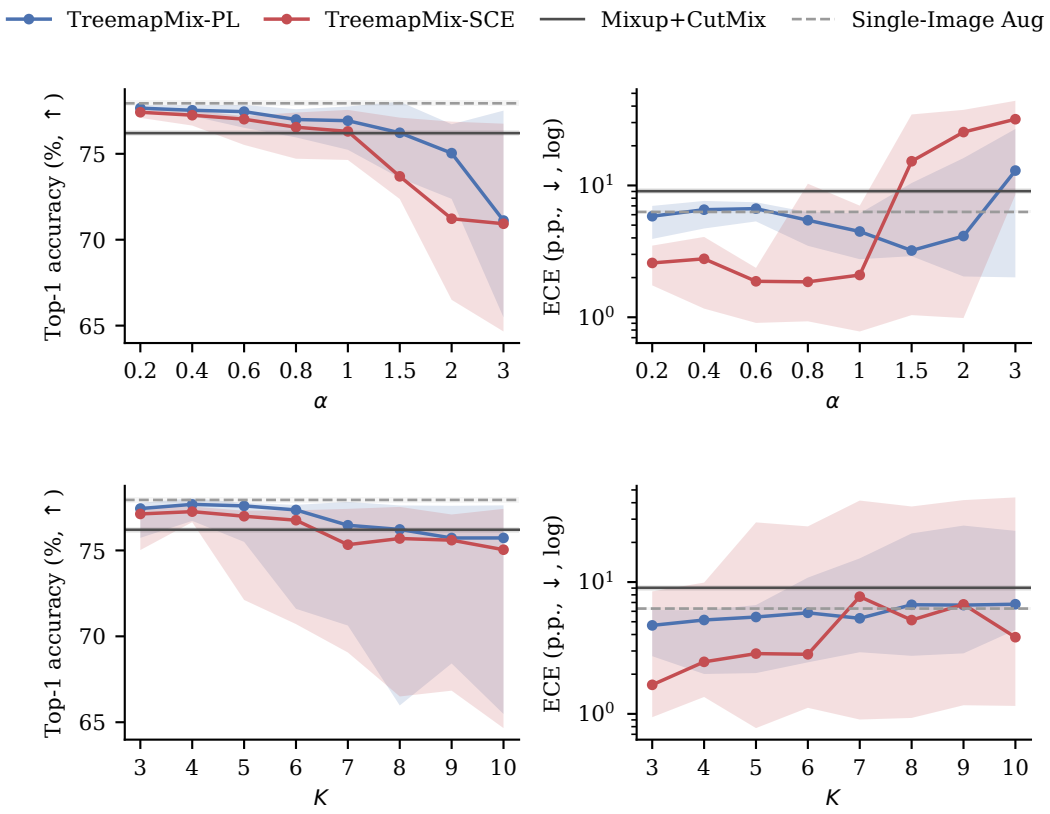


Figure 8: ImageNet-1K hyperparameter sensitivity of TreemapMix on ViT-Wee. The top row sweeps α and marginalizes over K, while the bottom row sweeps K and marginalizes over α. The displayed source data cover $K = 3, \ldots, 10$. Curves show the median across the marginalized hyperparameter, and shaded regions show the corresponding min–max range. The best accuracy–calibration trade-off occurs for small K and moderately small α. Full (K, α) heatmaps for Top-1 accuracy and ECE are shown in Appendix Figure 9. These experiments use the simplified sweep protocol without aspect-ratio filtering or K/α scheduling.

Figure 9 reports the available two-dimensional (K, α) heatmaps for the same sweep. These heatmaps show the interaction between the number of regions and the Dirichlet concentration, whereas Figure 8 summarizes the marginal trends.

For Top-1 accuracy, both TreemapMix-SCE and TreemapMix-PL perform best at relatively small α values and around $K = 4$. Increasing α up to approximately 1 causes only modest degradation, while larger values lead to a clearer accuracy drop. This trend suggests that TreemapMix benefits from some imbalance among region areas rather than from nearly uniform region sizes. Larger K values also tend to reduce Top-1 accuracy, but this effect is weaker and less uniform than the effect of α. As

K increases, performance varies more strongly across α values, indicating that high-composition settings are more sensitive to the choice of mixing strength.

Calibration, measured by ECE with 15 bins, follows a similar but sharper pattern. For both losses, ECE remains relatively stable up to a loss-dependent threshold and then increases substantially. For TreemapMix-SCE, this threshold occurs around $\alpha = 1$; for TreemapMix-PL, it occurs around $\alpha = 1.5$. Increasing K also tends to worsen calibration, but the effect is weaker and less systematic than the effect of α. Across the sweep, TreemapMix-PL is less sensitive to hyperparameter variation, while TreemapMix-SCE achieves the strongest calibration when α and K are chosen carefully.

These results indicate that choosing α is more important than choosing K for reliable TreemapMix performance. For calibration-focused use, the sweep supports TreemapMix-SCE with $\alpha \leq 1$ and $K \leq 5$. For accuracy-focused use, the sweep supports TreemapMix-PL with $\alpha \leq 0.4$ and $K \leq 6$. The main recipe filters layouts with extreme patch aspect ratios and schedules α from 0.1 to 0.5. These settings were not isolated in this sweep.

F.7 Qualitative Visualizations and Attribution Analysis

Figures 10, 11, and 12 visualize how TreemapMix changes with its random draw and main hyperparameters. Figure 10 holds the source images and hyperparameters fixed while varying only the random draw. Figures 11 and 12 sweep the number of regions and the Dirichlet concentration, respectively, showing how these controls change the granularity and balance of the mixed image.

We also complement the quantitative logit-alignment diagnostics with qualitative attribution maps in Figure 13. The maps are computed on the same held-out diagnostic set using the ViT-Betwixt checkpoints. We use Gradient$\times$Input attribution [37, 38], which multiplies the input by the gradient of the target logit with respect to the input. This provides a qualitative view of which source regions contribute most strongly to a given class logit.

Figure 13 shows different probability behavior across the losses. For boathouse, SCE assigns 0.01 while other methods round to 0.00; rounded zeros need not be exact zeros. PL assigns substantially higher probability to the largest region, consistent with ranking rather than probability matching.

TreemapMix attribution often concentrates on target regions, including the small suit and boathouse patches; baseline Cougar attributions focus more on the face. These qualitative observations complement the logit diagnostics rather than providing standalone evidence. PL shows the clearest ordinal correspondence.

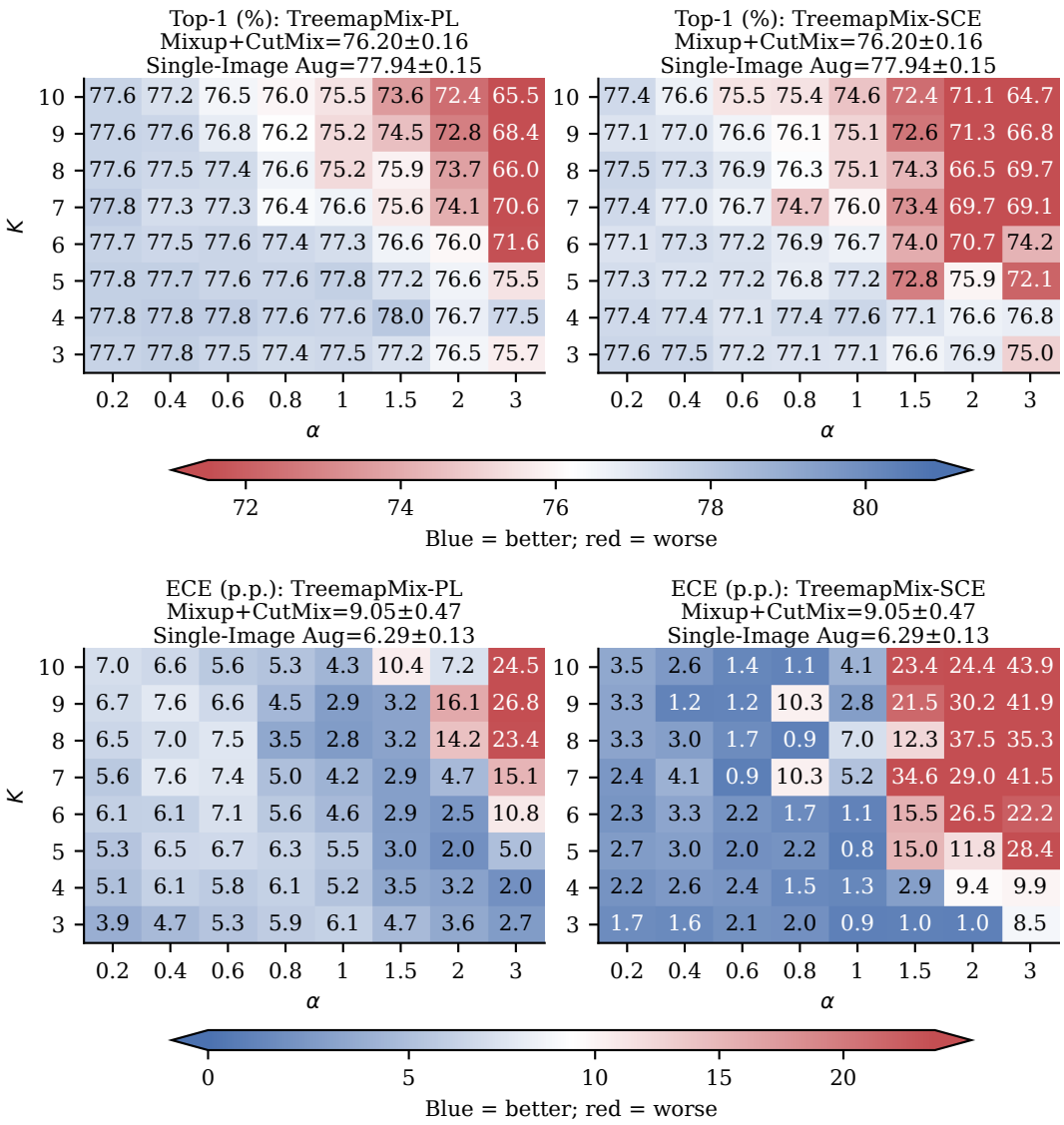


Figure 9: Appendix heatmaps for the ImageNet-1K TreemapMix hyperparameter sweep on ViT-Wee. ECE is measured using 15 bins. The top row shows Top-1 over the available (K, α) grid ($K = 3, \ldots, 10$), and the bottom row shows ECE over the same grid. Both plots use the same simplified sweep protocol as the line plots in Figure 8: we do not apply aspect filtering and we do not use α or K scheduling, so these absolute values are expected to be somewhat worse than the tuned main results. The heatmaps show that for best results in terms of ECE and accuracy, small K values together with moderately small α values should be used.



Figure 10: TreemapMix randomness for fixed sources, $\alpha = 0.5$, and $K = 5$. Independent Dirichlet draws produce different compositions while preserving sources and settings.

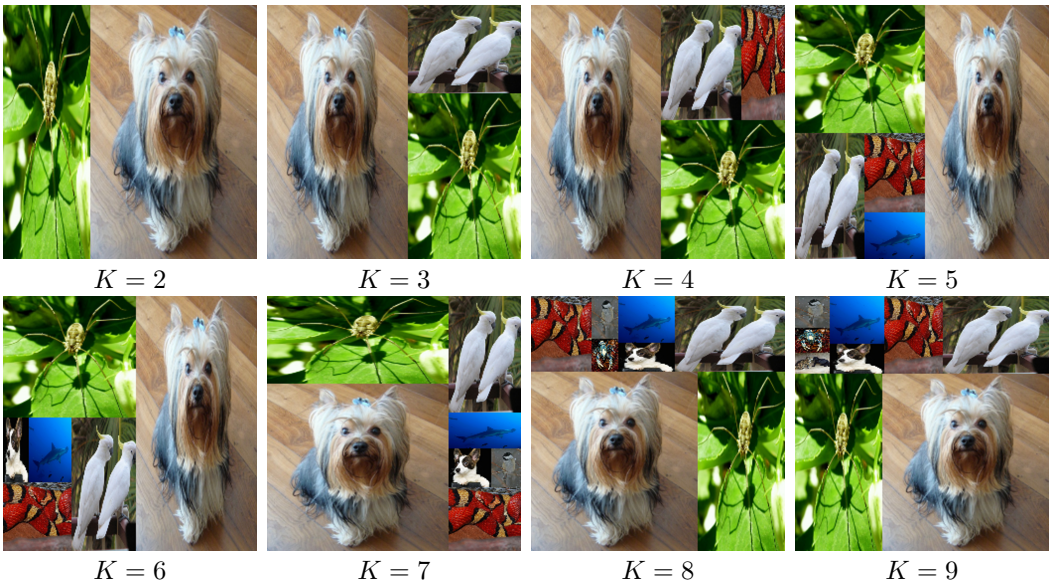


Figure 11: Effect of source count for fixed images, $\alpha = 2.0$, and maximum aspect ratio 15.0. Increasing K from 2 to 9 gives finer-grained mixtures.

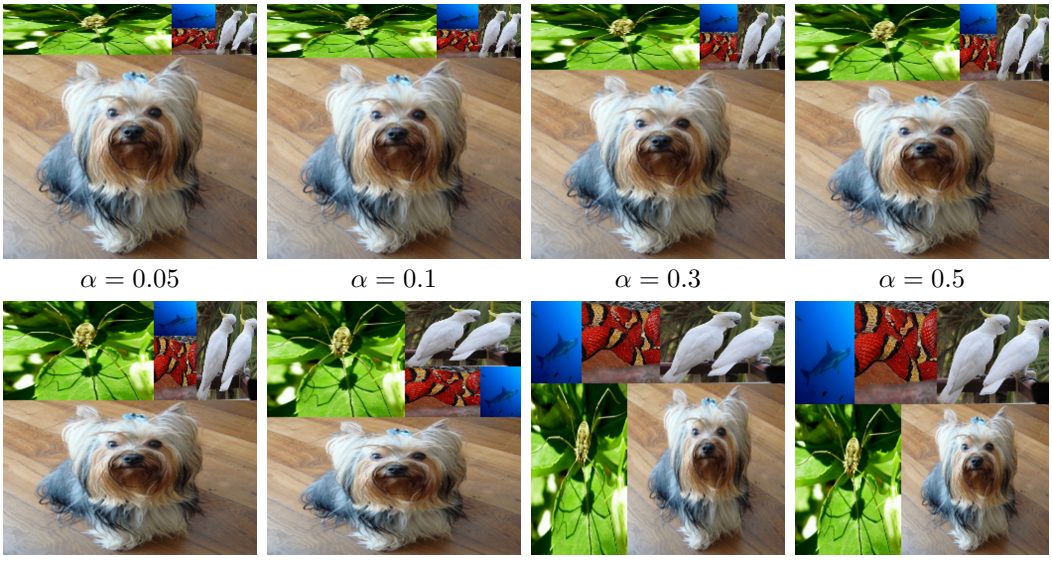


Figure 12: Effect of Dirichlet concentration for fixed images, $K = 5$, and maximum aspect ratio 15.0. Smaller α favors more unequal region areas.

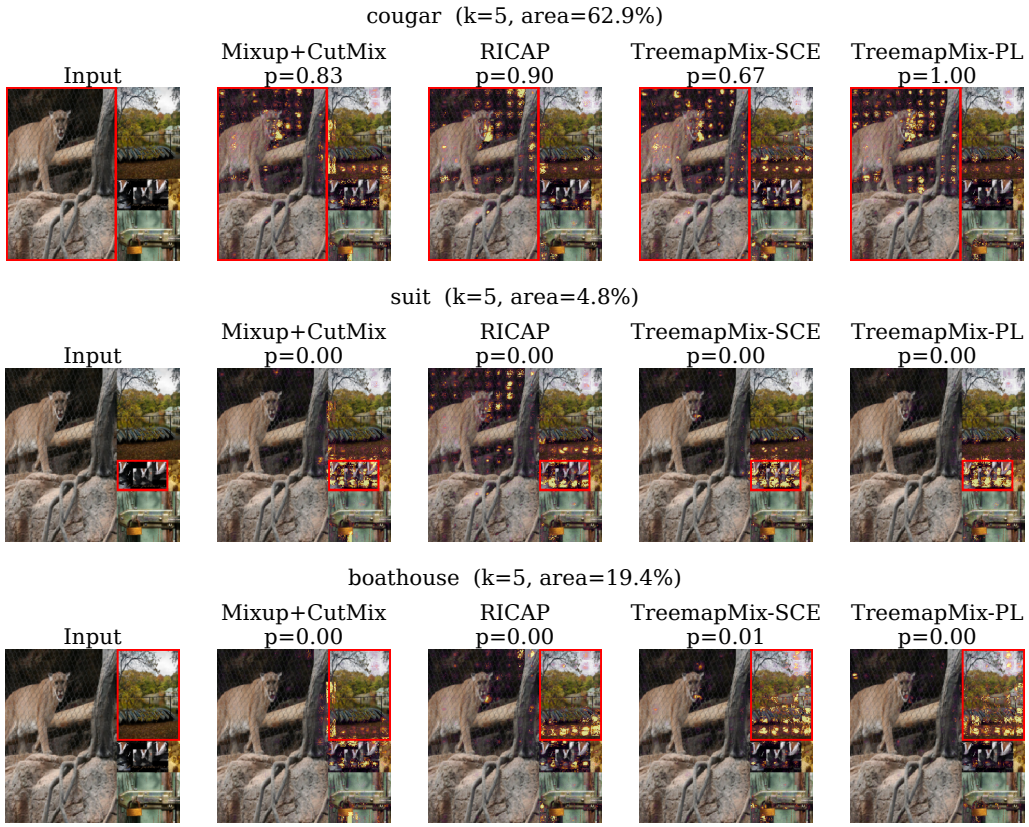


Figure 13: Gradient×Input attribution for three ImageNet-1K classes. Each row shows the boxed input and overlays for Mixup+CutMix, RICAP, SCE, and PL. Red boxes identify target patches; brighter regions indicate larger attribution. Labels give target probabilities and patch areas. TreemapMix often concentrates attribution on target regions; PL probabilities are sharper than SCE.

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